# MRK SIR SUGGESTION (AUTUMN 23)

Fourier series: Ex-22-25, 31, problem 19

Line spectrum: Ex-32, 33, 34

Convolution Sum, Convolution Integral: Follow Lecture Sheet

Laplace Transform: 58, 60, 61, 64, 68, 72, 75

Inverse Laplace Transform: Ex-101

Fourier transform: Ex-39-41, 43

Unit step function: ex-89, 91, 92

Ramp function: 90

Impulse function: Problem-31

Solve differential equation using Laplace Transform: ex-103, 104, 105

MATLAB

**Example 22**



$y = f(t) = 0; \quad -4 \le t \le 0$ ----------------------(i)

$y = 5; \quad 0 \le t \le 4$

$f(t) = f(t+8)$ **Here, $T = 2L = 8$** $\therefore L = 4$

a) Sketch the function for 3 cycles:
b) Find the Fourier series for the function

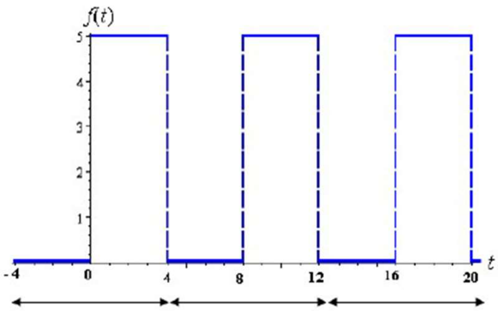



**Figure 55:** A periodic signal with period $T = 2L = 8$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(t)dt$$
$$= \frac{1}{4}\int_{-4}^{4} f(t)dt$$
$$= \frac{1}{4}\int_{-4}^{0} f(t)dt + \frac{1}{4}\int_{0}^{4} f(t)dt$$
$$= \frac{1}{4}\int_{-4}^{0} 0.dt + \frac{1}{4}\int_{0}^{4} 5dt \qquad \text{[From (i)]}$$
$$= 0 + \frac{1}{4}[5t]_0^4 \qquad [\int dt = t]$$
$$= \frac{1}{4}\times[5\times 4-0]$$

$$= \frac{1}{4} \times [5 \times 4 - 0]$$

$$= \frac{20}{4} = 5$$

$$a_n = \frac{1}{L} \int_{-L}^{L} f(t) \cos \frac{n\pi t}{L} dt$$



$$a_n = \frac{1}{4} \int_{-4}^{4} f(t) \cos \frac{n\pi t}{4} dt$$

$$a_n = \frac{1}{4} \int_{-4}^{0} f(t) \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_{0}^{4} f(t) \cos \frac{n\pi t}{4} dt$$

$$a_n = \frac{1}{4} \int_{-4}^{0} (0) \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_{0}^{4} (5) \cos \frac{n\pi t}{4} dt \qquad \text{[From (i)]}$$

$$a_n = 0 + \frac{1}{4} \times 5 \int_{0}^{4} \cos \frac{n\pi t}{4} dt$$

$$= \frac{5}{4} \times \frac{1}{\frac{n\pi}{4}} \left[ \sin \frac{n\pi t}{4} \right]_0^4 \qquad [\int \cos mx\, dx = \frac{1}{m} \sin mx]$$

$$= \frac{5}{4} \times \frac{4}{n\pi} \left[ \sin \frac{n\pi \times 4}{4} - \sin \frac{n\pi \times 0}{4} \right]$$

$$= \frac{5}{n\pi} [\sin n\pi - \sin 0]$$

$$= \frac{5}{n\pi} \sin n\pi$$

$$= 0 \qquad [\because \sin \pi = \sin 2\pi = \sin 3\pi = \dots\dots\dots = \sin n\pi = 0 \text{ and } \sin 0 = 0]$$

$$b_n = \frac{1}{L} \int_{-L}^{L} f(t) \sin \frac{n\pi t}{L} dt$$

$$b_n = \frac{1}{4} \int_{-4}^{4} f(t) \sin \frac{n\pi t}{4} dt$$

$$b_n = \frac{1}{4} \int_{-4}^{0} f(t) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_{0}^{4} f(t) \sin \frac{n\pi t}{4} dt$$

$$b_n = \frac{1}{4} \int_{-4}^{0} (0) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_{0}^{4} (5) \sin \frac{n\pi t}{4} dt \qquad \text{[From (i)]}$$

$$b_n = 0 + \frac{1}{4} \times 5 \int_{0}^{4} \sin \frac{n\pi t}{4} dt$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(t)\sin\frac{n\pi t}{L}dt$$

$$b_n = \frac{1}{4}\int_{-4}^{4} f(t)\sin\frac{n\pi t}{4}dt$$

$$b_n = \frac{1}{4}\int_{-4}^{0} f(t)\sin\frac{n\pi t}{4}dt + \frac{1}{4}\int_{0}^{4} f(t)\sin\frac{n\pi t}{4}dt$$

$$b_n = \frac{1}{4}\int_{-4}^{0} (0)\sin\frac{n\pi t}{4}dt + \frac{1}{4}\int_{0}^{4} (5)\sin\frac{n\pi t}{4}dt \qquad \text{[From (i)]}$$

$$b_n = 0 + \frac{1}{4}\times 5\int_{0}^{4}\sin\frac{n\pi t}{4}dt$$

$$b_n = \frac{5}{4}\int_{0}^{4}\sin\frac{n\pi t}{4}dt$$

$$= \frac{5}{4}\times\frac{1}{\frac{n\pi}{4}}\left[-\cos\frac{n\pi t}{4}\right]_0^4 = \frac{-5}{4}\times\frac{4}{n\pi}\left[\cos\frac{n\pi t}{4}\right]_0^4$$

$$[\int\cos mx\,dx = \frac{1}{m}\sin mx; \int\sin mx\,dx = \frac{-1}{m}\cos mx]$$

$$= \frac{-5}{n\pi}\left[\cos\frac{n\pi\times 4}{4} - \cos\frac{n\pi\times 0}{4}\right]$$



$$= \frac{-5}{n\pi}[\cos n\pi - \cos 0]$$

$$= \frac{-5}{n\pi}[\cos n\pi - 1]$$

The Fourier series for the above function:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi t}{L}$$

$$= \frac{5}{2} + \sum_{n=1}^{\infty}(0)\cos\frac{n\pi t}{L} + \sum_{n=1}^{\infty} -\frac{5}{n\pi}(\cos(n\pi) - 1)\sin\frac{n\pi t}{L}$$

$$= 2.5 - \frac{5}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}(\cos n\pi - 1)\sin\frac{n\pi t}{4}$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \underbrace{[-\frac{5}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}(\cos n\pi - 1)\sin\frac{n\pi t}{4}]}_{\text{AC value}}\ \text{Answer}$$

**Example 23:**

$$y = f(t) = -1\ ; \ -\pi < t < 0$$

$$= 1\ ; \ 0 < t < \pi \quad \text{---------------------(i)}$$

**Example 23:**

$y = f(t) = -1 \ ; \ -\pi < t < 0$ ----------------------(i)

$\quad\quad = \ 1 \ ; \quad 0 < t < \pi$

$f(t) = f(t + 2\pi)$ Here, $T = 2L = 2\pi \quad \therefore L = \pi$

a) Sketch the function for 3 cycles:

b) Find the Fourier series for the function

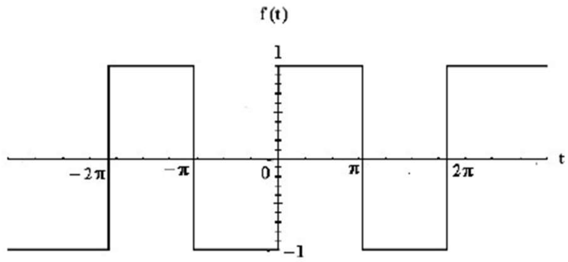



**Figure 56:** A periodic signal with period $T = 2L = 2\pi$

$$a_0 = \frac{1}{L}\int_{-\pi}^{\pi} f(t)dt$$



$$\Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} (-1)dt + \frac{1}{\pi}\int_{0}^{\pi} (1)dt \qquad \text{[From (i)]}$$

$$= \frac{-1}{\pi}[t]_{-\pi}^{0} + \frac{1}{\pi}[t]_{0}^{\pi}$$

$$= \frac{-1}{\pi}[0-(-\pi)] + \frac{1}{\pi}[\pi - 0]$$

$$= -\frac{1}{\pi}[\pi] + \frac{1}{\pi}[\pi]$$

$$\Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} (-1)dt + \frac{1}{\pi}\int_{0}^{\pi} (1)dt \qquad \text{[From (i)]}$$

$$= \frac{-1}{\pi}[t]_{-\pi}^{0} + \frac{1}{\pi}[t]_{0}^{\pi}$$

$$= \frac{-1}{\pi}[0-(-\pi)] + \frac{1}{\pi}[\pi - 0]$$

$$= -\frac{1}{\pi}[\pi] + \frac{1}{\pi}[\pi]$$

$$= -1 + 1$$

$$= 0$$

$$a_n = \frac{1}{L}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{L}dt$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)\cos\frac{n\pi t}{\pi} + \frac{1}{\pi}\int_{0}^{\pi} f(t)\cos\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} (-1)\cos ntdt + \frac{1}{\pi}\int_{0}^{\pi} (1)\cos ntdt \qquad \text{[From (i)]}$$

$$= -\frac{1}{\pi}\int_{-\pi}^{0} \cos ntdt + \frac{1}{\pi}\int_{0}^{\pi} \cos ntdt$$

$$= -\frac{1}{\pi}\times\frac{1}{n}[\sin nt]_{-\pi}^{0} + \frac{1}{\pi}\times\frac{1}{n}[\sin nt]_{0}^{\pi} \qquad [\int \cos mx\,dx = \frac{1}{m}\sin mx]$$

$$= -\frac{1}{\pi n}[\sin 0 - \sin(-n\pi)] + \frac{1}{\pi n}[\sin n\pi - 0]$$

$$= -\frac{1}{\pi n}[0 - \sin(-n\pi)] + \frac{1}{\pi n}[\sin n\pi - 0]$$

$$= -\frac{1}{\pi n}[0 + \sin(n\pi)] + \frac{1}{\pi n}[\sin n\pi - 0] \qquad [\because \sin(-\theta) = -\sin\theta]$$

$$= -\frac{1}{\pi n}[0+0] + \frac{1}{\pi n}[0-0]\ [\therefore \sin n\pi = 0 \text{ for } n = 1,2,3.............]$$

$$= 0$$

$$b_n = \frac{1}{L}\int_{-\pi}^{\pi} f(t)\sin\frac{n\pi t}{L}dt$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)\sin\frac{n\pi t}{\pi}dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)\sin\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)\sin ntdt + \frac{1}{\pi}\int_{0}^{\pi} f(t)\sin ntdt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} (-1)\sin ntdt + \frac{1}{\pi}\int_{0}^{\pi} (1)\sin ntdt \qquad \text{[From (i)]}$$

$$= -\frac{1}{\pi}\int_{-\pi}^{0} \sin ntdt + \frac{1}{\pi}\int_{0}^{\pi} \sin ntdt$$

$$= -\frac{1}{\pi}\times\frac{1}{n}[-\cos nt]_{-\pi}^{0} + \frac{1}{\pi}\times\frac{1}{n}[-\cos nt]_{0}^{\pi} \ [\because \int \sin mx\, dx = \frac{-1}{m}\cos mx]$$

$$= +\frac{1}{\pi n}[\cos nt]_{-\pi}^{0} + \frac{1}{\pi n}[-\cos nt]_{0}^{\pi}$$

$$= \frac{1}{\pi n}[\cos 0 - \cos(-n\pi)] - \frac{1}{\pi n}[\cos n\pi - \cos 0]$$

$$= \frac{1}{\pi n}[1 - \cos n\pi] - \frac{1}{\pi n}[\cos n\pi - 1] \qquad [\therefore \cos(-\theta) = \cos\theta]$$

$$= \frac{1}{\pi n} - \frac{1}{\pi n}\cos n\pi - \frac{1}{\pi n}\cos n\pi + \frac{1}{\pi n}$$

$$= \frac{2}{\pi n} - \frac{2}{\pi n}\cos n\pi$$

$$= \frac{2}{\pi n}(1 - \text{conn}\pi)$$

The Fourier series for the above function:

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi t}{L}$$

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi t}{\pi} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi t}{\pi}$$

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$$

$$\therefore f(t) = 0 + \sum_{n=1}^{\infty} 0.\cos nt + \sum_{n=1}^{\infty} \frac{2}{n\pi}(1 - \cos n\pi)\sin nt$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{0}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{2}{n\pi}(1 - \cos n\pi)\sin nt}_{\text{AC value}}$$

**Example 24:**

**Example 24:**



$$y = f(t) = -t\ ;\quad -\pi \le t \le 0$$

$$y = t\ ;\quad 0 \le t \le \pi \text{ ------------------------(i)}$$

$\mathbf{f(t) = f(t+2\pi)}$ Here, $\mathbf{T = 2L = 2\pi}$ $\quad \therefore \mathbf{L = \pi}$

a) Sketch the function for 3 cycles:
b) Find the Fourier series for the function

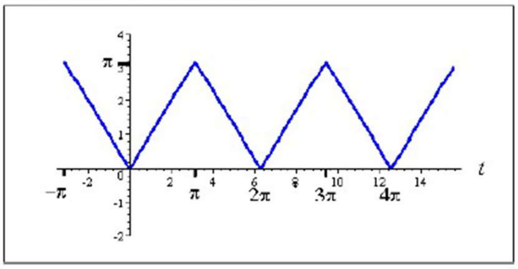



**Figure 57:** A periodic signal with period $\mathbf{T = 2L = 2\pi}$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(t)dt$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} (-t)dt + \frac{1}{\pi}\int_{0}^{\pi} t\,dt \qquad \text{[From (i)]}$$

$$= -\frac{1}{\pi}\left[\frac{t^2}{2}\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{t^2}{2}\right]_{0}^{\pi} \qquad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c\ ; n \ne -1\right]$$

$$= -\frac{1}{\pi}\left[0 - \frac{(-\pi)^2}{2}\right] + \frac{1}{\pi}\left[\frac{\pi^2}{2} - 0\right]$$

$$= -\frac{1}{\pi}\left[-\frac{\pi^2}{2}\right] + \frac{1}{\pi}\left[\frac{\pi^2}{2}\right]$$

$$= -\frac{1}{\pi}[-\frac{\pi^2}{2}] + \frac{1}{\pi}[\frac{\pi^2}{2}]$$
$$= \frac{1}{\pi}\times\frac{\pi^2}{2} + \frac{1}{\pi}\times\frac{\pi^2}{2}$$
$$= \frac{\pi}{2} + \frac{\pi}{2}$$



$$= \frac{2\pi}{2}$$
$$= \pi$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(t)\cos\frac{n\pi t}{L}dt$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{\pi}dt$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)dt$$
$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)\cos(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)\cos(nt)dt$$
$$= \frac{1}{\pi}\int_{-\pi}^{0} (-t)\cos(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} t\cos(nt)dt \qquad \text{[From (i)]}$$
$$= -\frac{1}{\pi}\int_{-\pi}^{0} t\cos(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} t\cos(nt)dt \text{ ------------------(ii)}$$

Now, Let, $\mathbf{I} = \int t\cos(nt)dt$

$$= t\int\cos(nt)dt - \int[\frac{d}{dt}(t)\int\cos(nt)dt]dt \quad [\because \int uvdx = u\int vdx - \int\{\frac{d}{dx}(u)\int vdx\}dx]$$
$$= t\frac{\sin(nt)}{n} - \int 1.\frac{\sin(nt)}{n}dt \qquad [\int\cos mx\,dx = \frac{1}{m}\sin mx]$$
$$= \frac{t}{n}\sin(nt) - \frac{1}{n}\int\sin(nt)dt$$
$$= \frac{t}{n}\sin(nt) - \frac{1}{n}\frac{(-\cos(nt))}{n} \qquad [\because \int\sin mx\,dx = \frac{-1}{m}\cos mx]$$
$$= \frac{t}{n}\sin(nt) + \frac{1}{n^2}\cos(nt) \text{ -------------------------(iii)}$$

Putting the value of (iii) in (ii)

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)dt = -\frac{1}{\pi}\int_{-\pi}^{0} t\cos(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} t\cos(nt)dt$$

$$= \frac{t}{n}\sin(nt) - \frac{1}{n}\frac{(-\cos(nt))}{n} \qquad [\because \int \sin mx\,dx = \frac{-1}{m}\cos mx]$$

$$= \frac{t}{n}\sin(nt) + \frac{1}{n^2}\cos(nt) \text{------------------------(iii)}$$

Putting the value of (iii) in (ii)

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)dt = -\frac{1}{\pi}\int_{-\pi}^{0} t\cos(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} t\cos(nt)dt$$

$$= -\frac{1}{\pi}\left[\frac{t}{n}\sin(nt) + \frac{1}{n^2}\cos(nt)\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{t}{n}\sin(nt) + \frac{1}{n^2}\cos(nt)\right]_{0}^{\pi}$$

$$= -\frac{1}{\pi}[\frac{0}{n}\sin(0) + \frac{1}{n^2}\cos(0) - \frac{(-\pi)}{n}\sin(-n\pi) - \frac{1}{n^2}\cos(-n\pi)] + \frac{1}{\pi}[\frac{\pi}{n}\sin(n\pi) + \frac{1}{n^2}\cos(n\pi)$$
$$- \frac{0}{n}\sin(0) - \frac{1}{n^2}\cos(0)]$$

$$= \frac{-1}{\pi}[0 + \frac{1}{n^2}\times 1 + \frac{\pi}{n}\sin(-n\pi) - \frac{1}{n^2}\cos(-n\pi)] + \frac{1}{\pi}[\frac{\pi}{n}\times 0 + \frac{1}{n^2}\cos(n\pi) - 0 - \frac{1}{n^2}\times 1]$$



$$= \frac{-1}{\pi}[\frac{1}{n^2} - \frac{\pi}{n}\sin(n\pi) - \frac{1}{n^2}\cos(n\pi)] + \frac{1}{\pi}[\frac{1}{n^2}\cos(n\pi) - \frac{1}{n^2}]$$

$[\sin(-\theta) = -\sin\theta\ ;\ \cos(-\theta) = \cos\theta;\ \sin n\pi = 0 \text{ for } n = 1,2,3..........]$

$$= \frac{-1}{\pi}[\frac{1}{n^2} - \frac{\pi}{n}\times 0 - \frac{1}{n^2}\cos(n\pi)] + \frac{1}{\pi}[\frac{1}{n^2}\cos(n\pi) - \frac{1}{n^2}]$$

$$= \frac{-1}{\pi n^2} + \frac{1}{\pi n^2}\cos(n\pi) + \frac{1}{\pi n^2}\cos(n\pi) - \frac{1}{\pi n^2}$$

$$= \frac{2}{\pi n^2}\cos n\pi - \frac{2}{\pi n^2}$$

$$= \frac{2}{\pi n^2}(\cos n\pi - 1) \qquad [\text{Taking Common } \frac{2}{\pi n^2}]$$

$$b_n = \frac{1}{L}\int_{-\pi}^{\pi} f(t)\sin\frac{n\pi t}{L}dt$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin\frac{n\pi t}{\pi}dt$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} f(t)\sin(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} f(t)\sin(nt)dt$$

$$= \frac{1}{\pi}\int (-t)\sin(nt)dt + \frac{1}{\pi}\int t\sin(nt)dt$$

Now $\int t\sin(nt)dt$

$$= t\int \sin(nt)dt - \int\{\frac{d}{dt}(t)\int \sin(nt)dt\}dt \quad [\because \int uvdx = u\int vdx - \int\{\frac{d}{dx}(u)\int vdx\}dx]$$

$$= t(\frac{-\cos(nt)}{n}) - \int 1.\frac{-\cos(nt)}{n}dt$$

$$= -\frac{t}{n}\cos(nt) + \frac{1}{n}\int \cos(nt)dt$$

$$= \frac{-t}{n}\cos(nt) + \frac{1}{n}\frac{\sin(nt)}{n} \qquad [\int \cos mx\, dx = \frac{1}{m}\sin mx]$$

$$= -\frac{t}{n}\cos(nt) + \frac{1}{n^2}\sin(nt) \text{-----------------------------(v)}$$

Putting the value of (v) in (iv)

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)dt = -\frac{1}{\pi}\int_{-\pi}^{0} t\sin(nt)dt + \frac{1}{\pi}\int_{0}^{\pi} t\sin(nt)dt$$



$$= -\frac{1}{\pi}\left[\frac{-t}{n}\cos(nt) + \frac{1}{n^2}\sin(nt)\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{-t}{n}\cos(nt) + \frac{1}{n^2}\sin(nt)\right]_{0}^{\pi}$$

$$= -\frac{1}{\pi}[\frac{-0}{n}\cos(0) + \frac{1}{n^2}\sin(0) - \frac{-(-\pi)}{n}\cos(-n\pi) - \frac{1}{n^2}\sin(-n\pi)] + \frac{1}{\pi}[-\frac{\pi}{n}\cos(n\pi)$$

$$+ \frac{1}{n^2}\sin(n\pi) - \frac{-(0)}{n}\cos(0) - \frac{1}{n^2}\sin(0)]$$

$$= -\frac{1}{\pi}[0 + 0 - \frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\sin(n\pi)] + \frac{1}{\pi}[-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\sin(n\pi) + 0 - 0]$$

$$= -\frac{1}{\pi}[-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\times 0] + \frac{1}{\pi}[-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\times 0]$$

$$[\sin(-\theta) = -\sin\theta \,;\ \cos(-\theta) = \cos\theta;\ \sin n\pi = 0 \text{ for } n = 1,2,3..........]$$

$$= \frac{\pi}{\pi n}\cancel{\cos}(n\pi) - \frac{\pi}{\pi n}\cancel{\cos}(n\pi)$$

$$= 0$$

The Fourier series for the above function is:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(\frac{n\pi}{L})t + \sum_{n=1}^{\infty} b_n\sin(\frac{n\pi}{L})t$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty}\frac{2}{n^2\pi}(\cos n\pi - 1)\cos(\frac{n\pi}{L})t + 0$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty}\frac{2}{n^2\pi}(\cos n\pi - 1)\cos(\frac{n\pi}{\pi})t$$

$$= -\frac{1}{\pi}\left[\frac{-t}{n}\cos(nt) + \frac{1}{n^2}\sin(nt)\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{-t}{n}\cos(nt) + \frac{1}{n^2}\sin(nt)\right]_{0}^{\pi}$$

$$= -\frac{1}{\pi}\left[\frac{-0}{n}\cos(0) + \frac{1}{n^2}\sin(0) - \frac{-(-\pi)}{n}\cos(-n\pi) - \frac{1}{n^2}\sin(-n\pi)\right] + \frac{1}{\pi}\left[-\frac{\pi}{n}\cos(n\pi)\right.$$

$$\left. + \frac{1}{n^2}\sin(n\pi) - \frac{-(0)}{n}\cos(0) - \frac{1}{n^2}\sin(0)\right]$$

$$= -\frac{1}{\pi}\left[0 + 0 - \frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\sin(n\pi)\right] + \frac{1}{\pi}\left[-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\sin(n\pi) + 0 - 0\right]$$

$$= -\frac{1}{\pi}\left[-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\times 0\right] + \frac{1}{\pi}\left[-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\times 0\right]$$

$[\sin(-\theta) = -\sin\theta\ ;\ \cos(-\theta) = \cos\theta;\ \sin n\pi = 0 \text{ for } n = 1,2,3..........]$

$$= \frac{\pi}{\pi n}\cancel{\cos}(n\pi) - \frac{\pi}{\pi n}\cancel{\cos}(n\pi)$$

$$= 0$$

The Fourier series for the above function is:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}\right)t + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}\right)t$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi}(\cos n\pi - 1)\cos\left(\frac{n\pi}{L}\right)t + 0$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi}(\cos n\pi - 1)\cos\left(\frac{n\pi}{\pi}\right)t$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{\frac{\pi}{2}}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{2}{n^2\pi}(\cos n\pi - 1)\cos nt}_{\text{AC value}}$$

**Example 25**

$y = f(t) = t\ ;\ -\pi \le t \le \pi$ -----------------------(i)

Here, $T = 2L = 2\pi$ $\qquad \therefore L = \pi$

a) Sketch the function for 3 cycles:

b) Find the Fourier series for the function

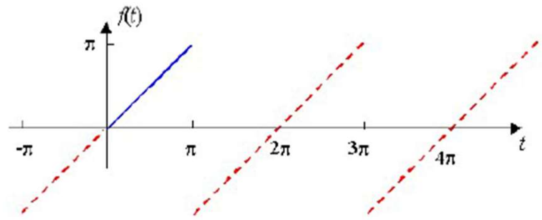


**Figure 58:** A periodic signal with period $\mathbf{T = 2L = 2\pi}$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(t)dt$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} t\,dt \qquad \text{[From (i)]}$$

$$= \frac{1}{\pi}\left[\frac{t^2}{2}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{\pi^2}{2} - \frac{(-\pi)^2}{2}\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi^2}{2} - \frac{\pi^2}{2}\right]$$

$$= 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{L}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{\pi}\,dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos nt dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} t\cos nt dt \qquad \text{[From (i)]}$$

Now $\int t\cos nt dt$

$$= t\int \cos nt dt - \int\{\frac{d}{dt}(t)\int \cos nt dt\}dt \;\; [\because \int uv dx = u\int v dx - \int\{\frac{d}{dx}(u)\int v dx\}dx]$$

$$= \frac{t}{n}\times \sin nt - \int 1.\frac{\sin nt}{n}dt$$

$$= \frac{t}{n}\sin nt - \frac{1}{n}.\frac{1}{n}.(-\cos nt)$$

$$= \frac{t}{n}\sin nt + \frac{1}{n^2}\cos(nt) \text{ ---------------------------(ii)}$$

$$\therefore a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t\cos nt dt$$

$$= \frac{1}{\pi}\left[\frac{t}{n}\sin nt + \frac{1}{n^2}\cos nt\right]_{-\pi}^{\pi} \qquad \text{[From (ii)]}$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n}\sin n\pi + \frac{1}{n^2}\cos n\pi - \left\{\frac{-\pi}{n}\sin(-n\pi) + \frac{1}{n^2}\cos(-n\pi)\right\}\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n}\sin n\pi + \frac{1}{n^2}\cos n\pi + \frac{\pi}{n}\sin(-n\pi) - \frac{1}{n^2}\cos(-n\pi)\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n}\sin n\pi + \frac{1}{n^2}\cos n\pi + \frac{\pi}{n}(-)\sin(n\pi) - \frac{1}{n^2}\cos(n\pi)\right]$$

$[\because \sin(-\theta) = -\sin\theta\ ,\ \cos(-\theta) = \cos\theta]$

$$= \frac{1}{\pi}\left[\frac{\pi}{n}\cancel{\sin} n\pi + \frac{1}{n^2}\cancel{\cos} n\pi - \frac{\pi}{n}\cancel{\sin} n\pi - \frac{1}{n^2}\cancel{\cos} n\pi\right]$$

$$= 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t).\sin\frac{n\pi t}{L}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin nt\,dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} t\sin nt\,dt \qquad \text{[From (i)]}$$

Now, $\int t\sin nt\,dt$

$$= t\int \sin nt\,dt - \int\{\frac{d}{dt}(t)\int \sin nt\,dt\}dt \quad [\because \int uv\,dx = u\int v\,dx - \int\{\frac{d}{dx}(u)\int v\,dx\}dx]$$

$$= t.\frac{1}{n}.(-\cos nt) - \int 1.\frac{1}{n}(-\cos nt)dt$$

$$= -\frac{t}{n}\cos nt + \frac{1}{n}\int \cos nt\,dt$$

$$= -\frac{t}{n}\cos nt + \frac{1}{n^2}\sin nt \text{ ---------------------------(iii)} \quad [\int \cos mx\,dx = \frac{1}{m}\sin mx]$$

$$\therefore b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t.\sin nt.dt$$

$$= \frac{1}{\pi}\left[-\frac{t}{n}\cos nt + \frac{1}{n^2}\sin nt\right]_{-\pi}^{\pi} \qquad \text{[From (iii)]}$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{n}\cos n\pi + \frac{1}{n^2}\sin n\pi - \left\{-\frac{-\pi}{n}\cos(-n\pi) + \frac{1}{n^2}\sin(-n\pi)\right\}\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{n}\cos n\pi + \frac{1}{n^2}\sin n\pi - \left\{\frac{\pi}{n}\cos(-n\pi) + \frac{1}{n^2}\sin(-n\pi)\right\}\right]$$

$$=\frac{1}{\pi}\left[-\frac{\pi}{n}\cos n\pi+\frac{1}{n^2}\sin n\pi-\frac{\pi}{n}\cos(-n\pi)-\frac{1}{n^2}\sin(-n\pi)\right]$$

$$=\frac{1}{\pi}\left[-\frac{\pi}{n}\cos n\pi+\frac{1}{n^2}\sin n\pi-\frac{\pi}{n}\cos(-n\pi)-\frac{1}{n^2}(-)\sin(n\pi)\right]$$

$$[\because \sin(-\theta)=-\sin\theta\ ;\cos(-\theta)=\cos\theta]$$

$$=\frac{1}{\pi}\left[-\frac{\pi}{n}\cos n\pi+\frac{1}{n^2}\sin n\pi-\frac{\pi}{n}\cos n\pi+\frac{1}{n^2}\sin n\pi\right]$$

$$=\frac{1}{\pi}\left[-\frac{\pi}{n}\cos n\pi-\frac{\pi}{n}\cos n\pi\right] \qquad [\because \sin n\pi = 0, \text{for } n = 1,2,3...]$$

$$=-\frac{1}{\pi}\times\frac{2\pi}{n}\cos n\pi$$

$$=-\frac{2}{n}\cos n\pi$$

The Fourier series for the above function is:

$$\therefore f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos nt+\sum_{n=1}^{\infty}b_n\sin nt$$

$$=0+0+\sum_{n=1}^{\infty}b_n\sin nt$$

$$=\sum_{n=1}^{\infty}-\frac{2}{n}\cos n\pi.\sin nt$$

$$\underbrace{f(t)}_{Complex\ wave}=\underbrace{0}_{DC\ value}+\underbrace{-2\sum_{n=1}^{\infty}\frac{1}{n}\cos n\pi.\sin nt}_{AC\ value}$$

**Example 31:**

**If $y = f(t) = t^2$ over the integral $-\pi < t < \pi$ and has period $2\pi$. Here, $T = 2L = 2\pi \therefore L = \pi$**

a) **Sketch three cycles of $y = f(t)$ in the interval $-3\pi < t < 3\pi$**
b) **Find the Fourier series for the function.**
c) *Hence, find the sum of the series* $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + - -$

**Answer:**



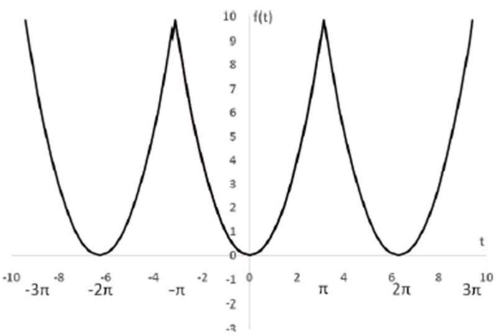



**Figure 60:** A periodic signal with period **T = 2L = 2π**

Let $y = f(t) = t^2$ --------------- (i)

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(t)dt$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)dt$$

**Figure 60:** A periodic signal with period $\mathbf{T = 2L = 2\pi}$

Let $y = f(t) = t^2$ --------------- (i)

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(t)dt$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)dt$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} t^2 dt \qquad \text{[From (i)]}$$

$$= \frac{1}{\pi}\left[\frac{t^3}{3}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{\pi^3}{3} - \frac{(-\pi)^3}{3}\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi^3}{3} + \frac{\pi^3}{3}\right]$$

$$= \frac{1}{\pi}\left[\frac{2\pi^3}{3}\right]$$

$$= \frac{2\pi^2}{3}$$



$$a_n = \frac{1}{L}\int_{-L}^{L} f(t)\cos\frac{n\pi t}{L}dt$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos nt dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} t^2 \cos nt dt \qquad \text{[From (i)]}$$

Now $\int t^2 \cos nt dt$

$$= t^2\int \cos nt dt - \int\{\frac{d}{dt}(t^2)\int \cos nt dt\}dt \; [\because \int uv dx = u\int v dx - \int\{\frac{d}{dx}(u)\int v dx\}dx]$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(t)\cos\frac{n\pi t}{L}\,dt$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos\frac{n\pi t}{\pi}\,dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos nt\,dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} t^2\cos nt\,dt \qquad \text{[From (i)]}$$

Now $\int t^2 \cos nt\,dt$

$$= t^2\int \cos nt\,dt - \int\{\frac{d}{dt}(t^2)\int \cos nt\,dt\}dt \quad [\because \int uv\,dx = u\int v\,dx - \int\{\frac{d}{dx}(u)\int v\,dx\}dx]$$

$$= \frac{t^2}{n}\sin nt - \int\{2t.\frac{\sin nt}{n}\}dt$$

$$= \frac{t^2}{n}\sin nt - \frac{2}{n}\left[t\int \sin nt - \int\{\frac{d}{dt}(t)\int \sin nt\,dt\}dt\right]$$

$$= \frac{t^2}{n}\sin nt - \frac{2}{n}\left[t.\frac{(-\cos nt)}{n} - \int\frac{(-\cos nt)}{n}dt\right]$$

$$= \frac{t^2}{n}\sin nt - \frac{2}{n}\left[\frac{-t\cos nt}{n} + \int\frac{\cos nt}{n}dt\right]$$

$$= \frac{t^2}{n}\sin nt - \frac{2}{n}\left[\frac{-t\cos nt}{n} + \frac{\sin nt}{n^2}\right]$$

$$= \frac{t^2}{n}\sin nt + \frac{2t}{n^2}\cos nt - \frac{2}{n^3}\sin nt$$

$$\int t^2\cos nt\,dt = \frac{t^2}{n}\sin nt + \frac{2t}{n^2}\cos nt - \frac{2}{n^3}\sin nt \text{ -------------- (ii)}$$

$$\therefore a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t^2\cos nt\,dt$$

$$= \frac{1}{\pi}\left[\frac{t^2}{n}\sin nt + \frac{2t}{n^2}\cos nt - \frac{2}{n^3}\sin nt\right]_{-\pi}^{\pi} \qquad \text{[From (ii)]}$$

$$= \frac{1}{\pi}\left[\frac{\pi^2}{n}\sin n\pi + \frac{2\pi}{n^2}\cos n\pi - \frac{2}{n^3}\sin n\pi - \frac{(-\pi)^2}{n}\sin n(-\pi) - \frac{2(-\pi)}{n^2}\cos n(-\pi) + \frac{2}{n^3}\sin n(-\pi)\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi^2}{n}\sin n\pi + \frac{2\pi}{n^2}\cos n\pi - \frac{2}{n^3}\sin n\pi + \frac{\pi^2}{n}\sin n\pi + \frac{2\pi}{n^2}\cos n\pi - \frac{2}{n^3}\sin n\pi\right]$$

$$[\sin(-\theta) = -\sin\theta;\ \cos(-\theta) = \cos\theta]$$

$$= \frac{1}{\pi}\left[0 + \frac{2\pi}{n^2}\cos n\pi - 0 + 0 + \frac{2\pi}{n^2}\cos n\pi - 0\right]$$

$$[\sin n\pi = 0; \text{For } n = 1,2,3.........]$$

$$= \frac{1}{\pi}\left[\frac{2\pi}{n^2}\cos n\pi + \frac{2\pi}{n^2}\cos n\pi\right]$$

$$= \frac{1}{\pi}\left[\frac{4\pi}{n^2}\cos n\pi\right]$$

$$= \frac{4}{n^2}\cos n\pi$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(t)\sin\frac{n\pi t}{L}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin\frac{n\pi t}{\pi}dt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin ntdt$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} t^2 \sin ntdt \qquad \text{[From (i)]}$$

Now $\int t^2 \sin ntdt$

$$= t^2\int \sin ntdt - \int\{\frac{d}{dt}(t^2)\int \sin ntdt\}dt \ [\because \int uvdx = u\int vdx - \int\{\frac{d}{dx}(u)\int vdx\}dx]$$

$$= t^2\int \sin ntdt - \int\{2t.\frac{1}{n}(-\cos nt)\}dt$$

$$= t^2\frac{1}{n}(-\cos nt) + \frac{2}{n}\int t\cos ntdt$$

$$= t^2\frac{1}{n}(-\cos nt) + \frac{2}{n}\left[t\int \cos nt - \int\{\frac{d}{dt}(t)\int \cos ntdt\}dt\right]$$

$$= -t^2\frac{1}{n}\cos nt + \frac{2}{n}\left[t.\frac{1}{n}\sin nt - \int 1.\frac{1}{n}\sin ntdt\right]$$

$$= -t^2\frac{1}{n}\cos nt + \frac{2}{n}\left[t.\frac{1}{n}\sin nt - \frac{1}{n}\int \sin ntdt\right]$$

$$= -t^2\frac{1}{n}\cos nt + \frac{2}{n}\left[t.\frac{1}{n}\sin nt - \frac{1}{n}.\frac{1}{n}(-\cos nt)\right]$$

$$= -t^2\frac{1}{n}\cos nt + \frac{2}{n}\left[t.\frac{1}{n}\sin nt + \frac{1}{n^2}\cos nt\right]$$

$$\int t^2 \sin nt dt = -t^2 \frac{1}{n}\cos nt + \frac{2t}{n^2}\sin nt + \frac{2}{n^3}\cos nt \text{ --------------(iii)}$$

$$\therefore b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t^2 \sin nt dt$$

$$= \frac{1}{\pi}\left[-t^2 \frac{1}{n}\cos nt + \frac{2t}{n^2}\sin nt + \frac{2}{n^3}\cos nt\right]_{-\pi}^{\pi} \qquad \text{[From (iii)]}$$

$$= \frac{1}{\pi}\left[-\frac{\pi^2}{n}\cos n\pi + \frac{2\pi}{n^2}\sin n\pi + \frac{2}{n^3}\cos n\pi - \frac{-(-\pi)^2}{n}\cos n(-\pi) - \frac{2(-\pi)}{n^2}\sin n(-\pi) - \frac{2}{n^3}\cos n(-\pi)\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi^2}{n}\cos n\pi + \frac{2\pi}{n^2}\sin n\pi + \frac{2}{n^3}\cos n\pi + \frac{\pi^2}{n}\cos n\pi - \frac{2\pi}{n^2}\sin n\pi - \frac{2}{n^3}\cos n\pi\right]$$

$$[\sin(-\theta) = -\sin\theta;\ \cos(-\theta) = \cos\theta]$$

$$= \frac{1}{\pi}\left[-\frac{\pi^2}{n}\cos n\pi + 0 + \frac{2}{n^3}\cos n\pi + \frac{\pi^2}{n}\cos n\pi - 0 - \frac{2}{n^3}\cos n\pi\right]$$

$$[\sin n\pi = 0; \text{For } n = 1,2,3\text{..........}]$$

$$= \frac{1}{\pi}\left[\cancel{-\frac{\pi^2}{n}\cos n\pi} + \cancel{\frac{2}{n^3}\cos n\pi} + \cancel{\frac{\pi^2}{n}\cos n\pi} - \cancel{\frac{2}{n^3}\cos n\pi}\right]$$

$$= \frac{1}{\pi}[0]$$

$$= 0$$

The Fourier series for the above function is:

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$$

$$= \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty}\frac{4}{n^2}\cos n\pi \cos nt + 0$$

$$= \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty}\frac{4}{n^2}\cos n\pi \cos nt$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{\frac{2\pi^2/3}{2}}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty}\frac{4}{n^2}\cos n\pi \cos nt}_{\text{AC value}} \qquad \text{Answer------------------(iv)}$$

$$f(t) = \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty}\frac{4}{n^2}\cos n\pi \cos nt$$

$$f(t) = \frac{\pi^2}{3} + \left(\frac{4}{1^2}\cos\pi\cos t + \frac{4}{2^2}\cos 2\pi\cos 2t + \frac{4}{3^2}\cos 3\pi\cos 3t + \frac{4}{4^2}\cos 4\pi\cos 4t + -----\right)$$

$$f(t) = \frac{\pi^2}{3} + \left(-4\cos t + \frac{4}{2^2}\cos 2t - \frac{4}{3^2}\cos 3t + \frac{4}{4^2}\cos 4t + -----\right)$$

$$f(t)=\frac{\pi^2}{3}-4(\cos t-\frac{1}{2^2}\cos 2t+\frac{1}{3^2}\cos 3t-\frac{1}{4^2}\cos 4t+-----)\text{--------------(v)}$$

**Given,** $y=f(t)=t^2\,;-\pi<t<\pi$

$$t^2=\frac{\pi^2}{3}-4(\cos t-\frac{1}{2^2}\cos 2t+\frac{1}{3^2}\cos 3t-\frac{1}{4^2}\cos 4t+-----)\text{---------------------(vi)}$$

Put $t=0$ in (vi)

$$0^2=\frac{\pi^2}{3}-4(\cos 0-\frac{1}{2^2}\cos 2.0+\frac{1}{3^2}\cos 3.0-\frac{1}{4^2}\cos 4.0+-----)$$

$$0^2=\frac{\pi^2}{3}-4(1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+-----)$$

$$-\frac{\pi^2}{3}=-4(1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+-----)$$

$$\frac{\pi^2}{3}=4(1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+-----)$$

$$\frac{\pi^2}{12}=(1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+-----)$$

$$1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+------=\frac{\pi^2}{12}$$

$$\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+------=\frac{\pi^2}{12}\ \text{Answer}$$

***Problem 19: Derive Complex form of Fourier series***

***We have, the trigonometric form of Fourier series is:***

$$f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos(n\omega t)+\sum_{n=1}^{\infty}b_n\sin(n\omega t)\text{----------------(iv)}$$

We have,

$$e^x=1+\frac{x^1}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\frac{x^5}{5!}+\frac{x^6}{6!}+\frac{x^7}{7!}+----------------$$

Put $x=ix$,

$$e^{ix}=1+\frac{ix}{1!}+\frac{(ix)^2}{2!}+\frac{(ix)^3}{3!}+\frac{(ix)^4}{4!}+\frac{(ix)^5}{5!}+\frac{(ix)^6}{6!}+\frac{(ix)^7}{7!}+--------$$

$[i^2=-1;\ i^3=i^2.i=-i;\ i^4=i^2.i^2=(-1).(-1)=+1;\ i^5=i^4.i=i]$

$$e^{ix}=1+\frac{ix}{1!}+\frac{-x^2}{2!}+\frac{-ix^3}{3!}+\frac{x^4}{4!}+\frac{ix^5}{5!}+\frac{-x^6}{6!}+\frac{-ix^7}{7!}+--------$$

$$e^{ix}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+-----+(\frac{ix^1}{1!}-\frac{ix^3}{3!}+\frac{ix^5}{5!}-\frac{ix^7}{7!}+--------)$$

$$e^{ix}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+-----+i(\frac{x^1}{1!}-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+--------)$$

$$[\because \cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+-----;\qquad \sin x=\frac{x^1}{1!}-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}---------]$$

$$\therefore e^{ix}=\cos x+i\sin x\text{-----------------------------------(v)}$$

Similarly,

$$e^x = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + ----------------$$

Put $x = -ix$,

$$e^{-ix} = 1 + \frac{-ix^1}{1!} + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \frac{(-ix)^5}{5!} + \frac{(-ix)^6}{6!} + \frac{(-ix)^7}{7!} + ----$$

$[(-i)^2 = -1; (-i)^3 = (-i)^2.(-i) = i; (-i)^4 = (-i)^2.(-i)^2 = (-1).(-1) = +1;$

$(-i)^5 = (-i)^4.(-i) = (+1).(-i) = -i]$

$$e^{-ix} = 1 + \frac{-ix^1}{1!} + \frac{-x^2}{2!} + \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{-ix^5}{5!} + \frac{-x^6}{6!} + \frac{ix^7}{7!} + ----$$

$$e^{-ix} = 1 - \frac{ix^1}{1!} - \frac{x^2}{2!} + \frac{ix^3}{3!} + \frac{x^4}{4!} - \frac{ix^5}{5!} - \frac{x^6}{6!} + \frac{ix^7}{7!} + ----$$

$$e^{-ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ---------- \frac{ix^1}{1!} + \frac{ix^3}{3!} - \frac{ix^5}{5!} + \frac{ix^7}{7!} + ----$$

$$e^{-ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + -------------- i(\frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ----)$$

$[\because \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + -----; \quad \sin x = \frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} ---------]$

$\therefore e^{-ix} = \cos x - i \sin x$ --------------------------------------(vi)

**Adding (v) and (vi),**

$e^{ix} = \cos x + i \sin x$

$e^{-ix} = \cos x - i \sin x$

$e^{ix} + e^{-ix} = 2\cos x$

$\therefore \cos x = \frac{1}{2}(e^{ix} + e^{-ix})$ ---------------------------------------(vii)

Again Subtracting (v) and (vi)

$e^{ix} = \cos x + i \sin x$

$e^{-ix} = \cos x - i \sin x$

$e^{ix} - e^{-ix} = 2i \sin x$

$\therefore \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$ ---------------------------------------(viii)

From equation (vii)

$\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$

Hence From equation (vii), we can write,

$$\therefore \cos(n\omega t) = \frac{1}{2}(e^{in\omega t} + e^{-in\omega t}) \text{ ------------------------------(ix)}$$

And from equation (vii)

$$\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$$

Hence from equation (vii), we can write

$$\therefore \sin(n\omega t) = \frac{1}{2i}(e^{in\omega t} - e^{-in\omega t}) \text{ --------------------------(x)}$$

Putting the values of (ix) and (x) in (iv), we get,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega t + \sum_{n=1}^{\infty} b_n \sin(n\omega t)$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \{\frac{1}{2}(e^{in\omega t} + e^{-in\omega t})\} + \sum_{n=1}^{\infty} b_n \{\frac{1}{2i}(e^{in\omega t} - e^{-in\omega t})\}$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[a_n \{\frac{1}{2}(e^{in\omega t} + e^{-in\omega t})\} + b_n \{\frac{1}{2i}(e^{in\omega t} - e^{-in\omega t})\}]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\{\frac{1}{2}(a_n e^{in\omega t} + a_n e^{-in\omega t})\} + \{\frac{1}{2i}(b_n e^{in\omega t} - b_n e^{-in\omega t})\}]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n e^{in\omega t} + a_n e^{-in\omega t}) + \frac{1}{2i}(b_n e^{in\omega t} - b_n e^{-in\omega t})]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n e^{in\omega t}) + \frac{1}{2i}(b_n e^{in\omega t}) + \frac{1}{2}(a_n e^{-in\omega t}) + \frac{1}{2i}(-b_n e^{-in\omega t})]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n e^{in\omega t}) + \frac{(-1)(-1)}{2i}(b_n e^{in\omega t}) + \frac{1}{2}(a_n e^{-in\omega t}) + \frac{(-1)(-1)}{2i}(-b_n e^{-in\omega t})]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n e^{in\omega t}) + \frac{(-1)(i^2)}{2i}(b_n e^{in\omega t}) + \frac{1}{2}(a_n e^{-in\omega t}) + \frac{(-1)(i^2)}{2i}(-b_n e^{-in\omega t})]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n e^{in\omega t}) + \frac{(-1)(i\ )}{2}(b_n e^{in\omega t}) + \frac{1}{2}(a_n e^{-in\omega t}) + \frac{(-1)(i\ )}{2}(-b_n e^{-in\omega t})]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n e^{in\omega t}) + \frac{-i}{2}(b_n e^{in\omega t}) + \frac{1}{2}(a_n e^{-in\omega t}) + \frac{-i}{2}(-b_n e^{-in\omega t})]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\{\frac{1}{2}(a_n e^{in\omega t}) - \frac{i}{2}(b_n e^{in\omega t})\} + \{\frac{1}{2}(a_n e^{-in\omega t}) + \frac{i}{2}(b_n e^{-in\omega t})\}]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n - ib_n)e^{in\omega t}) + \frac{1}{2}(a_n + ib_n)e^{-in\omega t})] \text{ ---------------------(xi)}$$

$$\text{Let, } c_0 = \frac{a_0}{2} \text{ ------------------------------------(a)}$$

$$c_n = \frac{1}{2}(a_n - ib_n) \text{ ------------------------------(b)}$$

$$c^*_n = \frac{1}{2}(a_n + ib_n) \text{ -----------------------------(c)}$$

Then from (viii), we get the series is:

$$f(t) = c_0 + \sum_{n=1}^{\infty}[c_n e^{in\omega t} + c^*_n e^{-in\omega t}]$$

$$f(t) = c_0 + \sum_{n=1}^{\infty}[c_n e^{in\omega t} + c_{-n} e^{-in\omega t}] \text{ [Say } c^*_n = c_{-n}]$$

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n e^{in\omega t} + \sum_{n=1}^{\infty} c_{-n} e^{-in\omega t} \text{ -------(d)= ---+------+.......}$$

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n e^{in\omega t} + \sum_{n=-1}^{-\infty} c_n e^{in\omega t} \text{ ---------(e)=----------+-------+-----}$$

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t} \text{ --------------(f)}$$

[We have, $\mathbf{c_n e^{in\omega t}}$

-1 0 1

Put $n = 0$, then

$\mathbf{c_n e^{in\omega t}}$

$= c_0 e^{i\times 0\times \omega t}$

$= c_0 e^0 = c_0.1 = c_0$]

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$ --------------------------------------*(xii) Which is referred to as the complex or exponential form of the Fourier Series expansion of the function f(t)*

Where

$\because \mathbf{e^{ix} = \cos x + i \sin x}$ [from (v)]

$\therefore \mathbf{e^{in\omega t} = \cos n\omega t + i \sin n\omega t}$ ------------------------------(xiii)

$$\mathbf{c_0 = \frac{a_0}{2} = \frac{1}{2}a_0 = \frac{1}{2}\frac{1}{L}\int_{-L}^{L} f(t)dt}$$

$$\mathbf{c_0 = \frac{1}{2L}\int_{-L}^{L} f(t)dt}$$

$\mathbf{c_0 = \frac{1}{T}\int_{-L}^{L} f(t)dt}$ **[Where Period T= 2L]----------------(xiv)**

and

$\mathbf{c_n = \frac{1}{2}(a_n - ib_n)}$ ------------------------------------------(xv)

$\mathbf{c^*_n = c_{-n} = \frac{1}{2}(a_n + ib_n)}$ --------------------------------(xvi)

We have,

$$\mathbf{a_n = \frac{1}{L}\int_{-L}^{L} f(t)\cos(n\omega t)dt}$$

$$\mathbf{b_n = \frac{1}{L}\int_{-L}^{L} f(t)\sin(nwt)dt}$$

$$\therefore c_n = \frac{1}{2}(a_n - ib_n)$$

$$= \frac{1}{2}\left(\frac{1}{L}\int_{-L}^{L} f(t)\cos(n\omega t)dt - i\frac{1}{L}\int_{-L}^{L} f(t)\sin(nwt)dt\right)$$

$$= \frac{1}{2L}\int_{-L}^{L}\{f(t)\cos(n\omega t) - i f(t)\sin(nwt)\}dt$$

$$= \frac{1}{2L}\int_{-L}^{L} f(t)\{\cos(n\omega t) - i\sin(nwt)\}dt$$

$$= \frac{1}{2L}\int_{-L}^{L} f(t)e^{-in\omega t}dt \text{ [Since } e^{-ix} = \cos x - i\sin x \text{ from (vi)]}$$

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(t)e^{-in\omega t}dt$$

$$c_n = \frac{1}{T}\int_{-L}^{L} f(t)e^{-in\omega t}dt \text{ [ T = 2L] ---------------------(xvii)}$$

and

$$c^*_n = c_{-n} = \frac{1}{2}(a_n + ib_n)$$

$$= \frac{1}{2}\left(\frac{1}{L}\int_{-L}^{L} f(t)\cos(n\omega t)dt + i\frac{1}{L}\int_{-L}^{L} f(t)\sin(nwt)dt\right)$$

$$= \frac{1}{2L}\left(\int_{-L}^{L} f(t)\cos(n\omega t)dt + i.\int_{-L}^{L} f(t)\sin(nwt)dt\right)$$

$$= \frac{1}{2L}\left(\int_{-L}^{L} f(t)\{\cos(n\omega t)dt + i\sin(nwt)dt\}\right)$$

$$= \frac{1}{2L}\int_{-L}^{L} f(t)e^{in\omega t}dt \text{ [}\because e^{ix} = \cos x + i\sin x \text{ from (2)]}$$

$$\therefore c^*_n = c_{-n} = \frac{1}{2L}\int_{-L}^{L} f(t)e^{in\omega t}dt$$

$$\therefore c^*_n = c_{-n} = \frac{1}{T}\int_{-L}^{L} f(t)e^{in\omega t}dt \text{ [Period T = 2L]-------(xviii)}$$

In summary, the complex form of the Fourier series expansion of a periodic function f (t), of period T, is:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

Where,

$$c_0 = \frac{1}{T}\int_{-L}^{L} f(t)dt$$



$$c_n = \frac{1}{T}\int_{-L}^{L} f(t)e^{-in\omega t}dt$$

$$c^*_n = c_{-n} = \frac{1}{T}\int_{-L}^{L} f(t)e^{in\omega t}dt$$

নতুন সিগন্যাল cos হবে যদি $\alpha = \frac{\pi}{2}$ হয়, যেখানে নতুন সিগন্যালের Amplitude $R = \sqrt{a^2 + b^2}$ ]

**Example 32:**

Let, $x = \frac{1}{4}\cos 8t + \frac{1}{8}\sin 8t$ -------------------------------------------(ix)

What is the amplitude of the resultant signal?

**Answer:**

We have, $a\cos\theta + b\sin\theta = R\sin(\theta + \alpha)$

So, in $x = \frac{1}{4}\cos 8t + \frac{1}{8}\sin 8t$

Here,

$a = \frac{1}{4}$ *and* $b = \frac{1}{8}$ *and* $\theta = 8$

(ix) নং সমীকরণ হতে, প্রথম সিগন্যালের সর্বোচ্চ Amplitude $\frac{1}{4}$ এবং দ্বিতীয় সিগন্যালের সর্বোচ্চ Amplitude $\frac{1}{8}$ । এই দুটি সিগন্যাল যোগ করলে যে নতুন সিগন্যাল পাওয়া যাবে তার সর্বোচ্চ Amplitude যদি R হয় তাহলে R এর মান বের করতে পারি যেখানে $R = \sqrt{(\frac{1}{4})^2 + (\frac{1}{8})^2}$ কারণ [$\because \mathbf{R} = \sqrt{\mathbf{a}^2 + \mathbf{b}^2}$]

$$\Rightarrow \mathbf{R} = \sqrt{(\frac{1}{4})^2 + (\frac{1}{8})^2}$$

$$\Rightarrow \mathbf{R} = \sqrt{\frac{1}{16} + \frac{1}{64}}$$



$$\Rightarrow \mathbf{R} = \sqrt{\frac{4+1}{64}}$$

$$\Rightarrow \mathbf{R} = \sqrt{\frac{5}{64}}$$

$$\Rightarrow \mathbf{R} = \frac{\sqrt{5}}{8} = 0.2795 \textit{Answer}$$

আর এই দুটি সিগন্যাল যোগ করলে সেই নতুন সিগন্যালটি sine এর আকার ধারণ করতে পারে আবার cosine এর আকার ধারণ করতে পারে । Remember that sine wave and cosine wave with Phase difference is $90^0$, Figure 19

**Figure 70: Frequency domain**

**Example 33:**
**A Fourier series is given below. List frequencies and plot the one-sided amplitude spectrum**

$$\underbrace{x(t)}_{\text{Complex wave}} = \underbrace{12}_{\text{DC value}} \underbrace{+9\cos(2\pi\times 10t+\frac{\pi}{3})+6\cos(2\pi\times 20t-\frac{\pi}{6})+4\cos(2\pi\times 30t+\frac{\pi}{4})}_{\text{AC value}}$$

$$[\because \omega = 2\pi f]$$

**Here DC value is 12; Amplitudes are 9, 6, and 4 with respect to frequencies 10, 20, 30 respectively where phases are $\frac{\pi}{3}, -\frac{\pi}{6}, \frac{\pi}{4}$**
**The frequencies are 0 (dc), 10 Hz, 20 Hz, 30 Hz**



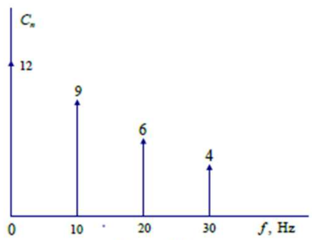


**Figure 71**

**Example 34:**

A plot showing each of the harmonic amplitudes in the wave is called the line spectrum.

**<u>Plot the line spectrum</u> (discrete frequency spectra) for the Fourier series:**

$$f(t)=\frac{\pi}{2}+\sum_{n=1}^{\infty}\frac{1}{2n-1}\cos nt+\sum_{n=1}^{\infty}\frac{(-1)^n}{2n}\sin nt \text{ ------------------------------------(i)}$$

This series has an interesting graph for the above function $\mathbf{f(t)}$

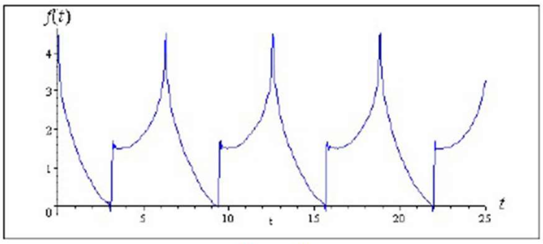



**Figure 77**



We have the Fourier series is $f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos(n\omega t)+b_n\sin(n\omega t))$

We can see from the series (i) that

$a_n=\frac{1}{2n-1} \qquad b_n=\frac{(-1)^n}{2n}$

Now, using $\Rightarrow R=\sqrt{a^2+b^2}$ [From (viii), page no 66]

Let $R_n=C_n=\sqrt{a_n^2+b_n^2}$

| $a_n=\frac{1}{2n-1}$ | $b_n=\frac{(-1)^n}{2n}$ | $C_n=\sqrt{a_n^2+b_n^2}$ |
|---|---|---|
| $a_1=1$ | $b_1=-\frac{1}{2}$ | $C_1=\sqrt{1^2+\left(-\frac{1}{2}\right)^2}=1.118$ |
| $a_2=\frac{1}{3}$ | $b_2=\frac{1}{4}$ | $C_2=\sqrt{\left(\frac{1}{3}\right)^2+\left(\frac{1}{4}\right)^2}=0.4167$ |
| $a_3=\frac{1}{5}$ | $b_3=-\frac{1}{6}$ | $C_3=\sqrt{\left(\frac{1}{5}\right)^2+\left(-\frac{1}{6}\right)^2}=0.260$ |

We have the Fourier series is $f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos(n\omega t)+b_n\sin(n\omega t))$

We can see from the series (i) that

$a_n = \frac{1}{2n-1} \qquad b_n = \frac{(-1)^n}{2n}$

Now, using $\Rightarrow R=\sqrt{a^2+b^2}$ [From (viii), page no 66]

Let $R_n = C_n = \sqrt{{a_n}^2+{b_n}^2}$

| $a_n = \frac{1}{2n-1}$ | $b_n = \frac{(-1)^n}{2n}$ | $C_n = \sqrt{a_n^2+b_n^2}$ |
|---|---|---|
| $a_1 = 1$ | $b_1 = -\frac{1}{2}$ | $C_1 = \sqrt{1^2+\left(-\frac{1}{2}\right)^2} = 1.118$ |
| $a_2 = \frac{1}{3}$ | $b_2 = \frac{1}{4}$ | $C_2 = \sqrt{\left(\frac{1}{3}\right)^2+\left(\frac{1}{4}\right)^2} = 0.4167$ |
| $a_3 = \frac{1}{5}$ | $b_3 = -\frac{1}{6}$ | $C_3 = \sqrt{\left(\frac{1}{5}\right)^2+\left(-\frac{1}{6}\right)^2} = 0.260$ |
| $a_4 = \frac{1}{7}$ | $b_4 = \frac{1}{8}$ | $C_4 = \sqrt{\left(\frac{1}{7}\right)^2+\left(\frac{1}{8}\right)^2} = 0.190$ |

Here, from (i),

*Here,* $n\omega = n$

**For $n = 1$;** **Fundamental Frequency = 1st Harmonic =** $\omega = 1$

**For $n = 2$;** **2nd Harmonic** $= 2\omega = 2$

**For $n = 3$;** **3rd Harmonic** $= 3\omega = 3$

**For $n = 4$;** **4th Harmonic** $= 4\omega = 4$



The resulting line spectrum is:

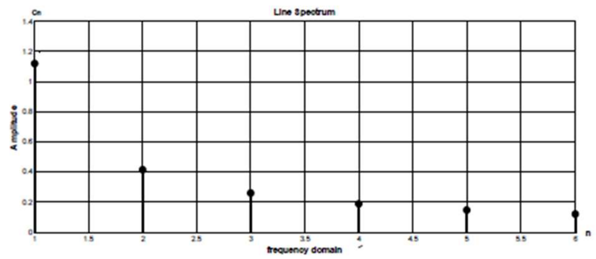



**Figure 78: Line spectrum**

## Convolution Integral

i. Folding
ii. Shifting
iii. Finding Area

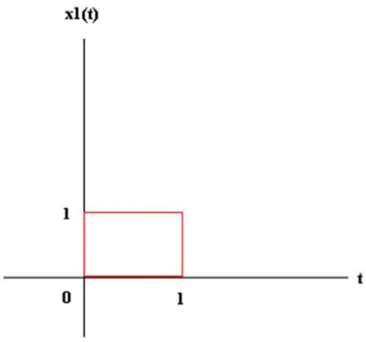
1st time:

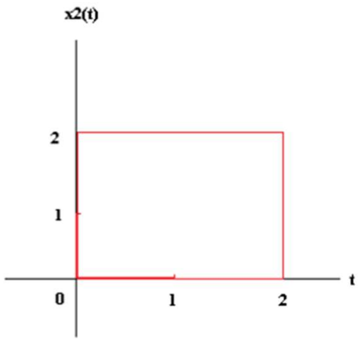
i. Folding
ii. Finding Area

2nd time and more

i. Shifting
ii. Finding Area

Q# 107 Find convolution integral of $x_1(t) * x_2(t)$



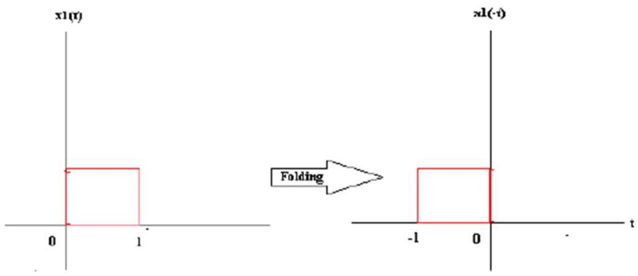




**1st time:**

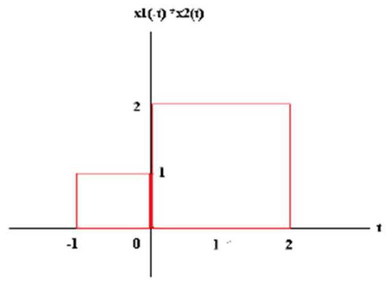
i. Folding

**1st time:**

i. Folding



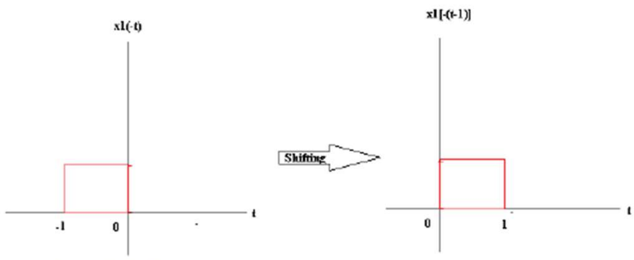
ii. Finding Area



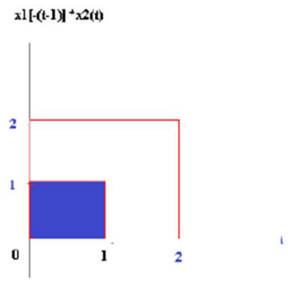

Finding Area: No overlapped area; x[0] =0

**2nd time:**

i. Shifting

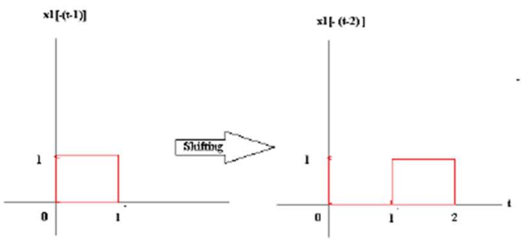



ii. Finding Area:

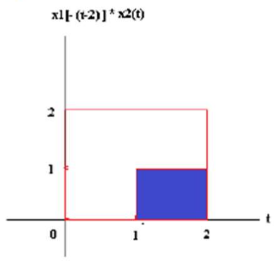



Area: x[1] $=1\times1=1$

iii. Shifting

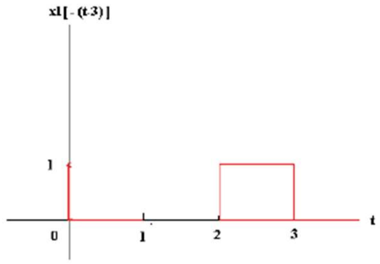



iv. Finding area

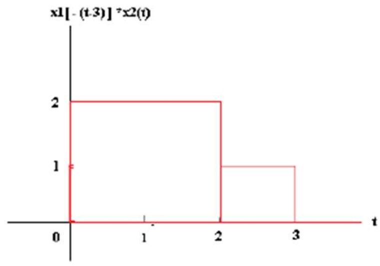


Area: $x[2] = 1 \times 1 = 1$

v. Shifting:

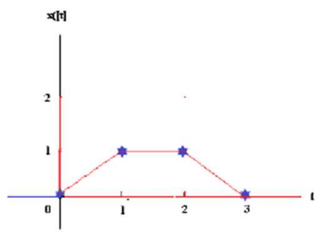



vi. Finding Area:




No overlapped: x [3] =0

Hence the convolution integral is



$\therefore L(f(t)) = L(t) = \frac{1}{s^2}$ *Answer*

**Example 58: Find Laplace Transform of $f(t) = e^{at}$**

***Answer:***

We have,

$$L(f(t)) = \int_0^\infty f(t)e^{-st}\,dt$$

Given,

$$f(t) = e^{at}$$

$$\therefore \; L(e^{at}) = \int_0^\infty e^{at}.e^{-st}\,dt$$

$$= \int_0^\infty e^{at-st}\,dt$$



$$= \int_0^\infty e^{-st+at}\,dt$$

$$= \int_0^\infty e^{-(st-at)}\,dt$$

$$= \int_0^\infty e^{-(s-a)t}\,dt$$

$$= \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty \qquad [\int e^{-mx}dx = \frac{e^{-mx}}{-m}]$$

$$= \frac{1}{-(s-a)}\left[e^{-\infty} - e^{-0}\right]$$

$$= -\frac{1}{(s-a)}\left[\frac{1}{e^\infty} - \frac{1}{e^0}\right]$$

$$= -\frac{1}{(s-a)}\left[\frac{1}{\infty} - \frac{1}{1}\right] \qquad [e^\infty = \infty]$$

$$= -\frac{1}{(s-a)}\left[0-1\right] \qquad \left[\frac{1}{\infty} = 0\right]$$

$$= \frac{1}{s-a} \; \textit{Answer}$$

$$\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a} \; \textit{Answer}$$

**Example 60: Find Laplace Transform of $f(t) = \sin at$**

*Answer:*

We have,

$$L(f(t)) = \int_0^\infty f(t)e^{-st}\,dt \text{ ------------------------------------(i)}$$

Given,

$$f(t) = \sin at$$



$$L(\sin \mathbf{at}) = \int_0^\infty \sin \mathbf{at}\; e^{-st}\,dt \text{ --------------------------------(ii)}$$

We have, $e^{i\theta} = \cos\theta + i\sin\theta$

$$\therefore \mathbf{e^{iat} = \cos at + i \sin at}$$

Here, $\cos at$ is the real $(\Re)$ part of $e^{iat}$ and $\sin at$ is the imaginary **(I)** part of $\mathbf{e^{iat}}$.

That is, $\cos at = \Re(e^{iat})$ and $\sin at = I(e^{iat})$

[$\Re(e^{iat})$ Means real part of $e^{iat}$ and $I(e^{iat})$ means imaginary part of $\mathbf{e^{iat}}$.]

We have, $\mathbf{L(f(t)) = \int_0^\infty f(t)e^{-st}dt}$

$$\Rightarrow L(f(t)) = L(\sin \mathbf{at}) = L\{I(e^{iat})\}$$

From (ii),

$$L(\sin \mathbf{at}) = \int_0^\infty \sin \mathbf{at}\; e^{-st}\,dt$$

$$= \int_0^\infty I\, e^{iat}.e^{-st}\,dt \qquad [\because \sin at = I(e^{iat})]$$

$$= I\int_0^\infty e^{iat}.e^{-st}\,dt \qquad \text{----------------------------(iii)}$$

$$= \mathbf{I}\int_0^\infty \mathbf{e^{(ia-s)t}dt}$$

$$= \mathbf{I}\int_0^\infty \mathbf{e^{-(s-ia)t}dt}$$

$$= I\left\{\left[\frac{e^{-(s-ia)t}}{-(s-ia)}\right]_0^\infty\right\} \qquad [\int e^{-mx}dx = \frac{e^{-mx}}{-m}]$$

$$= I\left\{\frac{e^{-(s-ia)\infty}}{-(s-ia)} - \frac{e^{-(s-ia)\cdot 0}}{-(s-ia)}\right\}$$

$$= I\left\{\left[\frac{e^{-(s-ia)t}}{-(s-ia)}\right]_0^{\infty}\right\} \qquad [\int e^{-mx}dx = \frac{e^{-mx}}{-m}]$$

$$= I\left\{\frac{e^{-(s-ia)\infty}}{-(s-ia)} - \frac{e^{-(s-ia)\cdot 0}}{-(s-ia)}\right\}$$

$$= I\left\{0 - \frac{e^{-0}}{-(s-ia)}\right\} \quad [\because e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0]$$

$$= I\left[\frac{1}{s-ia}\right] \qquad [\because e^{-0} = \frac{1}{e^{0}} = \frac{1}{1} = 1]$$

$$= I\left\{\frac{s+ia}{(s+ia)(s-ia)}\right\}$$ [Multiplying by $s+ia$ on numerator and denominator]

$$= I\left\{\frac{s+ia}{s^2 - i^2a^2}\right\}$$



$$= I\left\{\frac{s+ia}{s^2+a^2}\right\} \; [i^2 = -1]$$

$$= I\left\{\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}\right\} \text{------------------------(iv)}$$

Taking that imaginary part from (iv)

$\therefore L(f(t)) = L\ (\sin at) = \frac{a}{s^2+a^2}$ *Answer*

**Example 61: Find Laplace Transform of $f(t) = \cos at$**

***Answer:***

We have,

$$\mathbf{L(f(t)) = \int_0^{\infty} f(t)e^{-st}dt} \quad \text{----------------------------(i)}$$

Given,

$$f(t) = \cos at$$

$$L(\cos at) = \int_0^{\infty} \cos at\ e^{-st}\,dt \quad \text{--------------------(ii)}$$

We have, $\mathbf{e^{i\theta} = \cos\theta + i\sin\theta}$

$$\therefore \mathbf{e^{iat} = \cos at + i\sin at}$$

Here, $\cos at$ is the real part of $e^{iat}$ and $\sin at$ is the imaginary part of $\mathbf{e^{iat}}$.

That is, $\cos at = \Re(\mathbf{e^{iat}})$ and $\mathbf{\sin at = I(e^{iat})}$

$= I\left\{\frac{s+ia}{s^2+a^2}\right\}$ $[i^2 = -1]$

$= I\left\{\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}\right\}$ --------------------------(iv)

Taking that imaginary part from (iv)

$\therefore L(f(t)) = L\ (\sin at) = \frac{a}{s^2+a^2}$ *Answer*

**Example 61: Find Laplace Transform of $f(t) = \cos at$**

***Answer:***

We have,

$$\mathbf{L(f(t)) = \int_0^\infty f(t)e^{-st}dt} \quad \text{----------------------------(i)}$$

Given,

$$f(t) = \cos at$$

$$L(\cos at) = \int_0^\infty \cos at\ e^{-st}\,dt \quad \text{--------------------(ii)}$$

We have, $e^{i\theta} = \cos\theta + i\sin\theta$

$\therefore e^{iat} = \cos at + i\sin at$

Here, $\cos at$ is the real part of $e^{iat}$ and $\sin at$ is the imaginary part of $e^{iat}$.

That is, $\cos at = \Re(e^{iat})$ and $\sin at = I(e^{iat})$

[$\Re(e^{iat})$ Means real part of $e^{iat}$ and $I(e^{iat})$ means imaginary part of $e^{iat}$.]

We have, $\mathbf{L(f(t)) = \int_0^\infty f(t)e^{-st}dt}$

$\Rightarrow L(f(t)) = L(\cos at) = L\left\{\Re(e^{iat})\right\}$

From (ii),

$$L(\cos at) = \int_0^\infty \cos at\ e^{-st}\,dt$$

$$= \int_0^\infty \Re e^{iat}.e^{-st}\,dt \qquad [\because \cos at = \Re(e^{iat})]$$

$$= \Re\int_0^\infty e^{iat}.e^{-st}dt \quad \text{---------------------------(iii)}$$

$$= \Re\int_0^\infty e^{(ia-s)t}dt$$

$$= \Re\int_0^\infty e^{-(s-ia)t}dt$$

$$= \Re\left\{\left[\frac{e^{-(s-ia)t}}{-(s-ia)}\right]_0^\infty\right\} \qquad [\int e^{-mx}\,dx = \frac{e^{-mx}}{-m}]$$

$$= \Re\left\{\frac{e^{-(s-ia)\infty}}{-(s-ia)} - \frac{e^{-(s-ia)\cdot 0}}{-(s-ia)}\right\}$$

$$= \Re\left\{0 - \frac{e^{-0}}{-(s-ia)}\right\} \quad [\because e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0]$$

$$= \Re\left[\frac{1}{s-ia}\right] \qquad [\because e^{-0} = \frac{1}{e^{0}} = \frac{1}{1} = 1]$$

$$= \Re\left\{\frac{s+ia}{(s+ia)(s-ia)}\right\} \text{[Multiplying by } s+ia \text{ on numerator and denominator]}$$

$$= \Re\left\{\frac{s+ia}{s^2 - i^2a^2}\right\}$$

$$= \Re\left\{\frac{s+ia}{s^2+a^2}\right\} \ [i^2 = -1]$$

$$= \Re\left\{\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}\right\} \text{-------------------------(iv)}$$

Taking that real part from (iv)

$\therefore L(f(t)) = L\ (\cos at) = \frac{s}{s^2+a^2}$ *Answer*

As for example
We get,

$L\ (\sin at) = \frac{a}{s^2+a^2}$

$\therefore L\ (\sin 5t) = \frac{5}{s^2+5^2}$ *Answer*

and

$L\ (\cos at) = \frac{s}{s^2+a^2}$

$L\ (\cos 3t) = \frac{s}{s^2+3^2}$ *Answer*

**Example 62: Find Laplace Transform of $f(t) = \sinh at$**

Given $f(t) = \sinh at$

We have, $f(t) = \sinh at = \frac{1}{2}(e^{at} - e^{-at})$

**Example 64: Find $L\{e^{-at}t\} = ?$**

*Answer:*

Here, $f(t) = t$

The first shift theorem states that,

If $L\{f(t)\} = f(s)$ ---------------------------(i)

Then $L\{e^{-at}f(t)\} = f(s+a)$ ---------------------------(ii)

We have, according to equation no (i), $L(f(t)) = L(t) = \frac{1}{s^2}$ [Here $f(s) = \frac{1}{s^2}$]

If $f(s) = \frac{1}{s^2}$

$\therefore f(s+a) = \frac{1}{(s+a)^2}$

Hence, according to equation no (ii), we can write

$L\{e^{-at}t\} = f(s+a) = \frac{1}{(s+a)^2}$ *Answer*

$(s+3-4)$ $(s-1)$

**Example 68: Find $L\{e^{-3t} * e^{5t}\} = ?$**

Here, $f(t) = e^{5t}$

The first shift theorem states that,

If $L\{f(t)\} = f(s)$ ---------------------------(i)

Then $L\{e^{-3t}f(t)\} = f(s+3)$ ---------------------------(ii)

We have, according to equation no (i), $L(f(t)) = L(e^{5t}) = \frac{1}{(s-5)}$ [Here $f(s) = \frac{1}{(s-5)}$]



If $f(s) = \frac{1}{(s-5)}$

$\therefore f(s+3) = \frac{1}{(s+3-5)}$

Hence, according to equation no (ii), we can write

$L\{e^{-3t} * e^{5t}\} = f(s+3) = \frac{1}{(s+3-5)} = \frac{1}{(s-2)}$ *Answer*

**Example 72: Find $L\{t.e^{4t}\} = ?$**

Here, $f(t) = e^{4t}$

This theorem states that

If $L\{f(t)\} = f(s)$ ---------------------------(i)

Then, $L\{t.f(t)\} = -\frac{d}{ds}\{f(s)\}$ ---------------------------(ii)

We have, according to equation no (i), $L(f(t)) = L(e^{4t}) = \frac{1}{(s-4)}$ [Here $f(s) = \frac{1}{(s-4)}$]

Hence, according to equation no (ii), we can write

$$L\{t.f(t)\} = -\frac{d}{ds}\{f(s)\}$$

$$L\{t.e^{4t}\} = -\frac{d}{ds}\{f(s)\} \qquad [f(t) = e^{4t}]$$

$$L\{t.e^{4t}\} = -\frac{d}{ds}\{\frac{1}{(s-4)}\} \qquad [f(s) = \frac{1}{(s-4)}]$$

$$L\{t.e^{4t}\} = -\frac{d}{ds}\{(s-4)^{-1}\}$$

$$L\{t.e^{4t}\} = -(-1)(s-4)^{-1-1}\frac{d}{ds}(s-4)$$

$$L\{t.e^{4t}\} = (s-4)^{-2}(1-0)$$

$$L\{t.e^{4t}\} = (s-4)^{-2}.1$$

$$L\{t.e^{4t}\} = \frac{1}{(s-4)^2} \qquad \textit{Answer}$$

**Example 75: Find Laplace Transform of $\{\frac{\sin 2t}{t}\}$**

**Answer:**

Here, $f(t) = \sin 2t$

Division theorem states that,

If $L\{f(t)\} = f(s)$ ---------------------------(i)



Then $L\left[\frac{1}{t}f(t)\right] = \int_s^\infty f(s)\,ds$ ---------------------------(ii)

We have, according to equation no (i), $L(f(t)) = L(\sin 2t) = \frac{2}{s^2+4}$ [Here $f(s) = \frac{2}{s^2+4}$]

Hence, according to equation no (ii), we can write

$$L\left[\frac{1}{t}f(t)\right] = \int_s^\infty f(s)\,ds$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = \int_s^\infty f(s)\,ds$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = \int_s^\infty \frac{2}{s^2+4}\,ds$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = \int_s^\infty \frac{2}{s^2+2^2}\,ds$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = 2\int_s^\infty \frac{1}{s^2+2^2}\,ds$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = 2\times\frac{1}{2}\left[\tan^{-1}\frac{s}{2}\right]_s^\infty$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = 2\times\frac{1}{2}\left[\tan^{-1}\frac{\infty}{2} - \tan^{-1}\frac{s}{2}\right]$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = 2\times\frac{1}{2}\left[\tan^{-1}\infty - \tan^{-1}\frac{s}{2}\right]$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = 2\times\frac{1}{2}\left[\tan^{-1}\tan\frac{\pi}{2} - \tan^{-1}\frac{s}{2}\right] \qquad [\because \tan\frac{\pi}{2} = \infty]$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = 2\times\frac{1}{2}\left[\frac{\pi}{2} - \tan^{-1}\frac{s}{2}\right]$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = \left[\frac{\pi}{2} - \tan^{-1}\frac{s}{2}\right]$$

$$L\left[\frac{1}{t}f(t)\right] = L\left[\frac{1}{t}\sin 2t\right] = \left[\frac{\pi}{2} - \tan^{-1}\frac{s}{2}\right] \qquad \textit{Answer}$$

$\left(s^2 + 5^2\right)$

**Example 101:**

Find inverse Laplace transform of: $\dfrac{s+4}{s(s-1)(s-2)}$

Solution:

Let,

$$\frac{s+4}{s(s-1)(s-2)} = \frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s-2)} \text{ ........................ } (i)$$

Multiplying by $s(s-1)(s-2)$ in both sides

$$\Rightarrow \frac{s+4}{s(s-1)(s-2)} \times s(s-1)(s-2) = A\frac{s(s-1)(s-2)}{s} + B\frac{s(s-1)(s-2)}{(s-1)} + C\frac{s(s-1)(s-2)}{(s-2)}$$

$$\Rightarrow s+4 = A(s-1)(s-2) + Bs(s-2) + Cs(s-1) \quad \text{.............. } (ii)$$

Put $s = 0$ in equation $(ii)$,

$\Rightarrow 0+4 = A(0-1)(0-2) + B \times 0(0-2) + C \times 0(0-1)$

$\Rightarrow 4 = 2A$

$\therefore A = 2$

Put $s-1=0$, i.e. $s=1$ in equation $(ii)$,

$\Rightarrow 1+4 = A(1-1)(1-2) + B\times 1(1-2) + C\times 1(1-1)$

$\Rightarrow 5 = 0 - B + 0$

$\therefore B = -5$

Put $s-2=0$, i.e. $s=2$ in equation $(ii)$,

$\Rightarrow 2+4 = A(2-1)(2-2) + B\times 2(2-2) + C\times 2(2-1)$

$\Rightarrow 6 = 0+0+C(4-2)$

$\Rightarrow 6 = 0+0+2C$

$\therefore C = 3$

Putting the value of A, B, C in equation $(i)$, we get,

$$\frac{s+4}{s(s-1)(s-2)} = \frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s-2)}$$

$$\frac{s+4}{s(s-1)(s-2)} = \frac{2}{s} + \frac{-5}{(s-1)} + \frac{3}{(s-2)}$$

$$\therefore L^{-1}\left(\frac{s+4}{s(s-1)(s-2)}\right) = L^{-1}\left(\frac{2}{s}\right) + L^{-1}\left(\frac{-5}{s-1}\right) + L^{-1}\left(\frac{3}{s-2}\right)$$

$$L^{-1}\left(\frac{s+4}{s(s-1)(s-2)}\right) = 2L^{-1}\left(\frac{1}{s}\right) - 5L^{-1}\left(\frac{1}{s-1}\right) + 3L^{-1}\left(\frac{1}{s-2}\right) \text{---------}(iii)$$

Since

01. We have $\therefore L(f(t)) = L(1) = \frac{1}{s}$ [Example 55]

$\therefore 1 = L^{-1}(\frac{1}{s})$

$\therefore L^{-1}(\frac{1}{s}) = 1$

02. We have, $\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a}$ [Example 58]

$\therefore e^{at} = L^{-1}(\frac{1}{s-a})$

$\therefore e^{t} = L^{-1}(\frac{1}{s-1})$

$\therefore L^{-1}(\frac{1}{s-1}) = e^{t}$



03. $\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a}$ [Example 58]

$\therefore e^{at} = L^{-1}(\frac{1}{s-a})$

$\therefore e^{2t} = L^{-1}(\frac{1}{s-2})$

$\therefore L^{-1}(\frac{1}{s-2}) = e^{2t}$ Answer

Putting these values in (iii), we get

$$L^{-1}\left(\frac{s+4}{s(s-1)(s-2)}\right) = 2L^{-1}\left(\frac{1}{s}\right) - 5L^{-1}\left(\frac{1}{s-1}\right) + 3L^{-1}\left(\frac{1}{s-2}\right)$$

$$= 2.1 - 5e^{t} + 3e^{2t} \text{ Answer}$$

## Convolution Sum

The following steps are to be taken

i. Folding
ii. Shifting
iii. Multiplication
iv. Summation

**1st times:**

i. Folding
ii. Multiplication
iii. Summation

**2nd times and more**

i. Shifting
ii. Multiplication
iii. Summation

**Example 102:**

Evaluate the convolution sums of y [n] = x [n]*h [n]

Where,



$$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases} \quad \text{and} \quad h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$$ ; *n represents the time index*

*MATLAB*

*x=[1 3]*

*h=[2 1]*

*y =conv(x,h)*

$$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases} \quad \text{and} \quad h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$$ ; *n represents the time index*

*MATLAB*

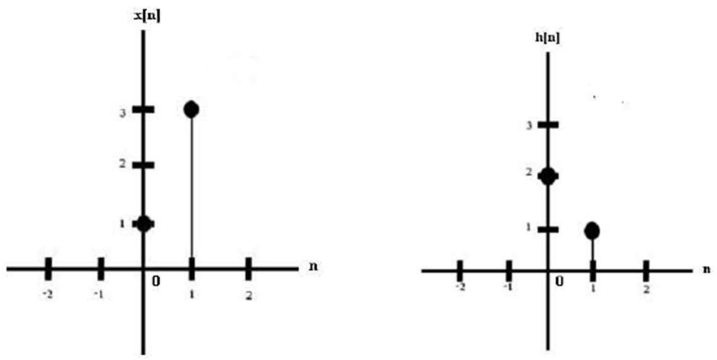
*x=[1 3]*

*h=[2 1]*

*y =conv(x,h)*

*y= [2 7 3]*

**Solution:**



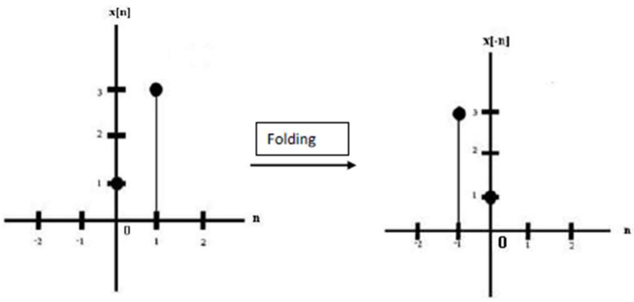

**1st time:**

**(i). Folding:**

$$x[n] = x[-n]$$

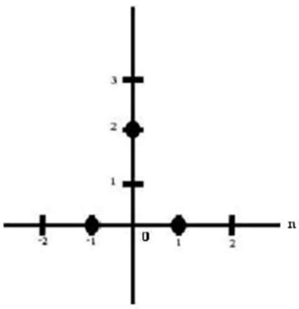
i.e. $x[0] = x[0]$

$x[1] = x[-1]$




**(ii). Multiplication:**

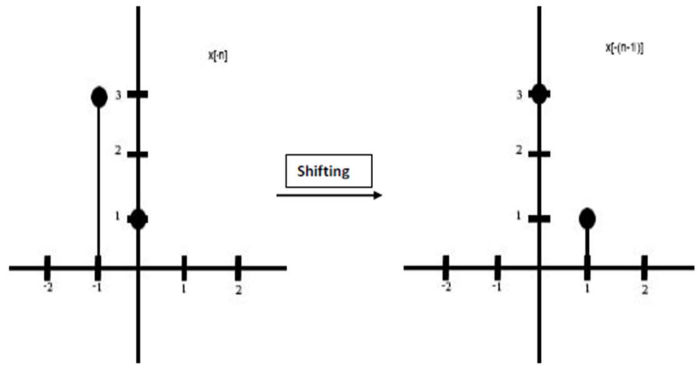
x[-n] * h[n]



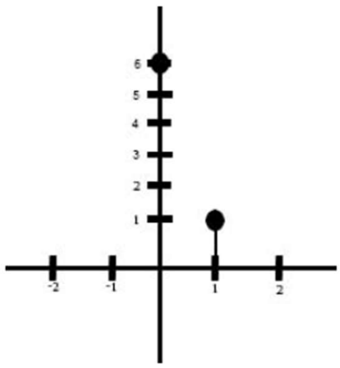
**(iii). Summation**: y[0] = 0 + 2 + 0 = 2

**2nd time:**

**(i). Shifting:**



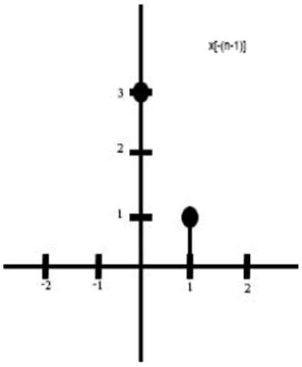

**(ii) Multiplication:** x[-(n-1)] * h[n]

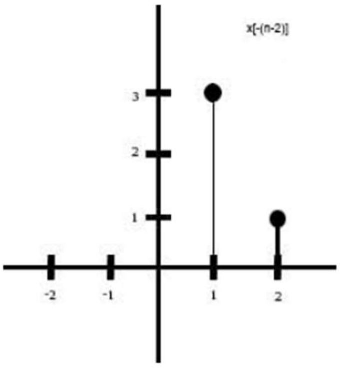


**(iii). Summation:** $y[1] = 0 + 6 + 1 + 0 = 7$

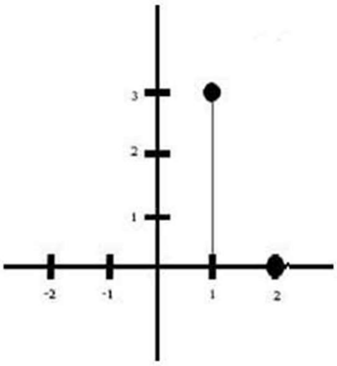
**3rd time:**

**(i). Shifting:**



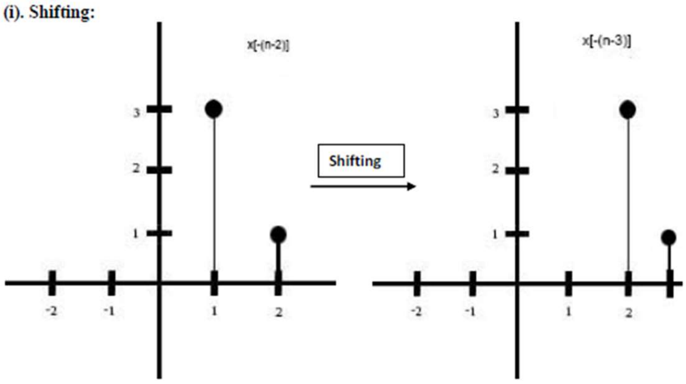

**(ii). Multiplication:** x[-(n-2)] * h[n]



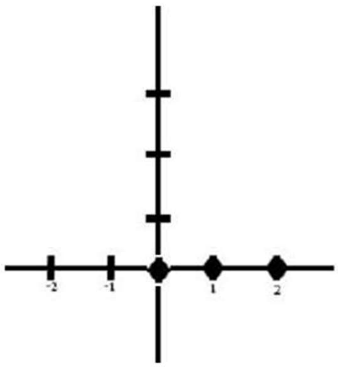

**(iii). Summation:** y [2] = 0 + 0 + 3 + 0 = 3

**$4^{th}$ time:**

**(i). Shifting:**




**(ii). Multiplication:** x[-(n-3)] * h[n]




iii) Summation: y[3] = 0+0+0+0=0

Finally we get,

∴ y[0] = 2

y[1] = 7

y[2] = 3

**∴ Convolution Sum: y[n] = [2 7 3]**

**Example 39: Consider the rectangular pulse figure 84 described as**

$\mathbf{f(t) = 1; \qquad -T \le t \le T}$

$\mathbf{\quad = 0; \qquad |t| > T}$

Find the Fourier Transform of $\mathbf{f(t)}$

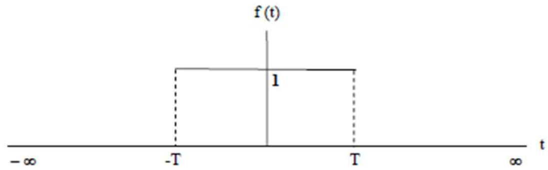



**Figure 88:** Rectangular Pulse

**Answer:**

We have,

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}[\ g(\omega)]\,e^{i\omega t}\,d\omega$$

Where $g(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$

$$g(\omega) = \int_{-\infty}^{-T} f(t)e^{-i\omega t}dt + \int_{-T}^{T} f(t)e^{-i\omega t}dt + \int_{T}^{\infty} f(t)e^{-i\omega t}dt$$

$$g(\omega) = \int_{-\infty}^{-T} 0.e^{-i\omega t}dt + \int_{-T}^{T} 1.e^{-i\omega t}dt + \int_{T}^{\infty} 0.e^{-i\omega t}dt$$

$$g(\omega) = \int_{-T}^{T} 1.e^{-i\omega t}dt = [\frac{e^{-i\omega t}}{-i\omega}]_{-T}^{T} \qquad [\because \int e^{-mx}dx = \frac{e^{-mx}}{-m}]$$

$$= \frac{-1}{i\omega}(e^{-i\omega T} - e^{i\omega T}) = \frac{1}{i\omega}(e^{i\omega T} - e^{-i\omega T})$$

$$g(\omega) = \frac{2.1}{\omega}\{\frac{1}{2i}(e^{i\omega T} - e^{-i\omega T})\} = \frac{2}{\omega}\sin(\omega T) \text{------------------------}(i)$$

$[\because \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$, **page no 84, equation no viii]**

OR

[Since we have, $\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$ ]

$$g(\omega) = \frac{2.1}{\omega}\{\frac{1}{2i}(e^{i\omega T} - e^{-i\omega T})\}$$

$$= \frac{2}{\omega}\sin(\omega T)$$

$$= \frac{2T}{\omega T}\sin(\omega T)$$

$$g(\omega) = 2T\frac{\sin(\omega T)}{\omega T}$$

Thus we write for all $\omega$ , $g(\omega) = 2T\frac{\sin(\omega T)}{\omega T}$ -----------------------------------(ii)

Both (i) & (ii) are correct

For $\omega = 0$, the integral simplifies to 2T. It is straightforward to show using L' Hopital's rule that, from (i)

$$\lim_{\omega \to 0}\frac{2}{\omega}\sin(\omega T)$$

$$= \lim_{\omega \to 0}\frac{2\sin(\omega T)}{\omega}[\frac{0}{0}; \text{Indeterminate Form}]$$

$$= \lim_{\omega \to 0}\frac{2\cos(\omega T).T}{1} [\text{Differentiate numerator and denominator with respect to } \omega]$$

$$= \frac{2\cos 0.T}{1}$$

$$= 2.1.T = 2T$$

$$\therefore \lim_{\omega \to 0}\frac{2}{\omega}\sin(\omega T) = 2T$$

Again, From (ii),

$$g(\omega) = 2T\frac{\sin(\omega T)}{\omega T}$$

$$\lim_{\omega \to 0} g(\omega) = \lim_{\omega \to 0} 2T\frac{\sin(\omega T)}{\omega T}$$

$$\lim_{\omega \to 0} g(\omega) = 2T\lim_{\omega \to 0}\frac{\sin(\omega T)}{\omega T}$$

$$\lim_{\omega \to 0} g(\omega) = 2T.1 \qquad \left[\because \lim_{\theta \to 0}\frac{\sin\theta}{\theta} = 1\right]$$

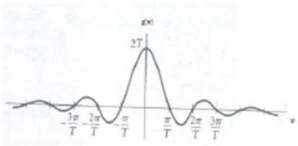



**Figure 89**: Fourier Transform of f (t)

**Example 40: Find the Fourier Transform of $f(t) = e^{-at}u(t)$**

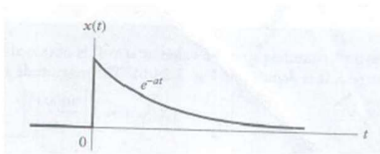



**Figure 90**: Exponential Signal

*Answer*: The Fourier Transform does not converge for $a \leq 0$ , since f (t) is not absolutely integrable, as shown by:

i. **For a>0, we have,**

$$g(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$

$$g(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-i\omega t}dt$$

$$g(\omega) = \int_{-\infty}^{0} e^{-at}u(t)e^{-i\omega t}dt + \int_{0}^{\infty} e^{-at}u(t)e^{-i\omega t}dt$$

$$g(\omega) = \int_{-\infty}^{0} e^{-at}.0.e^{-i\omega t}dt + \int_{0}^{\infty} e^{-at}.1.e^{-i\omega t}dt \qquad \text{[By definition from unit step function]}$$

$$g(\omega) = 0 + \int_{0}^{\infty} e^{-at}.1.e^{-i\omega t}dt$$

$$g(\omega) = \int_{0}^{\infty} e^{-at}e^{-i\omega t}dt$$

$$g(\omega)=\int_0^\infty e^{-at}e^{-i\omega t}dt=\int_0^\infty e^{-(a+i\omega)t}dt \text{ ---------------------------------(i)}$$

$$=\frac{-1}{a+i\omega}\left[e^{-(a+i\omega)t}\right]_0^\infty$$

$$g(\omega)=\frac{-1}{a+i\omega}[e^{-(a+i\omega)\infty}-e^{-(a+i\omega)0}]$$

$$g(\omega)=\frac{-1}{a+i\omega}[e^{-\infty}-e^{-0}]$$

$$=\frac{-1}{a+i\omega}[\frac{1}{e^\infty}-\frac{1}{e^0}]=\frac{-1}{a+i\omega}\left[\frac{1}{\infty}-\frac{1}{1}\right]=\frac{-1}{a+i\omega}[0-1]=\frac{1}{a+i\omega}\quad [\because e^\infty=\infty]$$

$$|g(\omega)|=\frac{1}{\sqrt{a^2+\omega^2}}$$

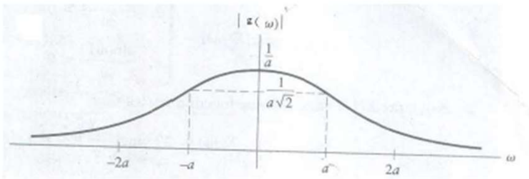



**Figure 91**: Fourier Transform of f (t) or Magnitude Spectrum

**ii. for a < 0**

$$g(\omega)=\int_0^\infty e^{-(a+i\omega)t}dt$$

$$g(\omega)=\int_0^\infty e^{-at-i\omega t}dt$$

$$g(\omega)=\int_0^\infty e^{at-i\omega t}dt \qquad [\text{for } a<0]$$

$$g(\omega)=\int_0^\infty e^{(a-i\omega)t}dt$$

$$g(\omega)=\frac{1}{a-i\omega}\left[e^{(a-i\omega)t}\right]_0^\infty$$

$$g(\omega)=\frac{1}{a-i\omega}\left[e^{(a-i\omega)\infty}-e^{(a-i\omega)0}\right]_0^\infty$$

$$g(\omega)=\frac{1}{a-i\omega}\left[e^{\infty}-e^{0}\right]$$

$$g(\omega)=\frac{1}{a-i\omega}[\infty-1]=\infty$$

**Example 41: Find Fourier Transform of**

$$f(t)=1 \qquad ;0\le t<1$$
$$=-1 \qquad ;-1\le t<0$$
$$=0 \qquad ;|t|>1$$

We have $g(\omega)=\int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,dt$

$$g(\omega)=\int_{-\infty}^{-1} f(t)e^{-i\omega t}\,dt+\int_{-1}^{0} f(t)e^{-i\omega t}\,dt+\int_{0}^{1} f(t)e^{-i\omega t}\,dt+\int_{1}^{\infty} f(t)e^{-i\omega t}\,dt$$

$$g(\omega)=\int_{-\infty}^{-1} 0.e^{-i\omega t}\,dt+\int_{-1}^{0} (-1)e^{-i\omega t}\,dt+\int_{0}^{1} 1.e^{-i\omega t}\,dt+\int_{1}^{\infty} 0.e^{-i\omega t}\,dt$$

$$g(\omega)=-\int_{-1}^{0} e^{-i\omega t}\,dt+\int_{0}^{1} e^{-i\omega t}\,dt$$

$$g(\omega)=-\left[\frac{e^{-i\omega t}}{-i\omega}\right]_{-1}^{0}+\left[\frac{e^{-i\omega t}}{-i\omega}\right]_{0}^{1} \qquad [\because \int e^{-mx}dx=\frac{e^{-mx}}{-m}]$$

$$g(\omega)=-\left[\frac{e^{-i\omega.0}}{-i\omega}-\frac{e^{-i\omega(-1)}}{-i\omega}\right]+\left[\frac{e^{-i\omega.1}}{-i\omega}-\frac{e^{-i\omega.0}}{-i\omega}\right]$$

$$g(\omega)=-\left[\frac{e^{-0}}{-i\omega}-\frac{e^{i\omega}}{-i\omega}\right]+\left[\frac{e^{-i\omega}}{-i\omega}-\frac{e^{-0}}{-i\omega}\right]$$

$$g(\omega)=-\left[\frac{\frac{1}{e^{0}}}{-i\omega}-\frac{e^{i\omega}}{-i\omega}\right]+\left[\frac{e^{-i\omega}}{-i\omega}-\frac{\frac{1}{e^{0}}}{-i\omega}\right]$$

$$g(\omega)=-\left[\frac{\frac{1}{1}}{-i\omega}-\frac{e^{i\omega}}{-i\omega}\right]+\left[\frac{e^{-i\omega}}{-i\omega}-\frac{\frac{1}{1}}{-i\omega}\right]$$

$$g(\omega)=-\left[\frac{1}{-i\omega}-\frac{e^{i\omega}}{-i\omega}\right]+\left[\frac{e^{-i\omega}}{-i\omega}-\frac{1}{-i\omega}\right]$$

$$g(\omega)=\left[\frac{1}{i\omega}-\frac{e^{i\omega}}{i\omega}\right]+\left[-\frac{e^{-i\omega}}{i\omega}+\frac{1}{i\omega}\right]$$

$$g(\omega)=\frac{1}{i\omega}+\frac{1}{i\omega}-\frac{e^{i\omega}}{i\omega}-\frac{e^{-i\omega}}{i\omega}$$



$$g(\omega)=\frac{2}{i\omega}-\frac{1}{i\omega}(e^{i\omega}+e^{-i\omega})$$

$$g(\omega)=\frac{2}{i\omega}-\frac{1}{i\omega}\frac{2}{2}(e^{i\omega}+e^{-i\omega})$$

$$g(\omega)=\frac{2}{i\omega}-\frac{2}{i\omega}\frac{1}{2}(e^{i\omega}+e^{-i\omega})$$

$$g(\omega)=\frac{2}{i\omega}-\frac{2}{i\omega}\cos\omega \qquad [\because \cos x=\frac{1}{2}(e^{ix}+e^{-ix})]$$

$$g(\omega)=\frac{2}{i\omega}(1-\cos\omega)\ \textit{Answer}$$

*Answer*

**Example 43: Find Fourier Transform for the given functions**

$f(t) = e^{-2t}$ ;$t \ge 0$

$f(t) = 0$ ;$t < 0$

*Answer:*

**Given**

$f(t) = e^{-2t}$ ;$t \ge 0$

$f(t) = 0$ ;$t < 0$ --------------------------(i)

**We have,**

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^{0} f(t) e^{-i\omega t} dt + \int_{0}^{\infty} f(t) e^{-i\omega t} dt$$



$$g(\omega) = \int_{-\infty}^{0} 0.e^{-i\omega t} dt + \int_{0}^{\infty} e^{-2t} e^{-i\omega t} dt$$ [Given equation no (i)]

$$g(\omega) = \int_{0}^{\infty} e^{-2t} e^{-i\omega t} dt$$

$$g(\omega) = \int_{0}^{\infty} e^{-2t-i\omega t} dt$$

$$g(\omega) = \int_{0}^{\infty} e^{-(2+i\omega)t} dt$$

$$g(\omega) = \left[\frac{e^{-(2+i\omega)t}}{-(2+i\omega)}\right]_0^{\infty}$$

$$g(\omega) = \frac{-1}{2+i\omega}\left[e^{-\infty} - e^{-0}\right]$$

$$g(\omega) = \frac{-1}{2+i\omega}\left[\frac{1}{e^{\infty}} - \frac{1}{e^{0}}\right]$$

$$g(\omega) = \frac{-1}{2+i\omega}\left[\frac{1}{\infty} - \frac{1}{1}\right]$$

$$g(\omega) = \frac{-1}{2+i\omega}\left[0-1\right]$$

$$g(\omega) = \frac{1}{2+i\omega}$$

$$|\,g(\omega)| = \frac{1}{\sqrt{2^2+\omega^2}}$$

$$|\,g(\omega)| = \frac{1}{\sqrt{4+\omega^2}}$$ *Answer*

**Example 89: Given that,** $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

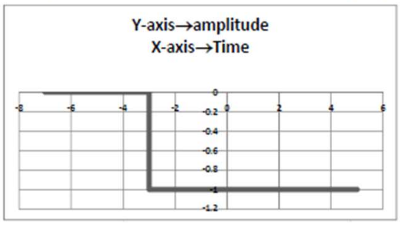
**Answer:**

**01.** $-u(t+3)$ =>

So,

$-u(t+3) = -1\,;\ t \geq -3;$ $\qquad$ $here, t+3 = 0$

$= 0;\ \ t < -3$ $\qquad$ $\therefore t = -3$

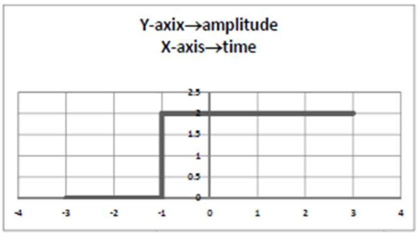



**02.** $2u(t+1)$



$2u(t+1) = 2; t \geq -1$ $\qquad$ $here, t+1 = 0$

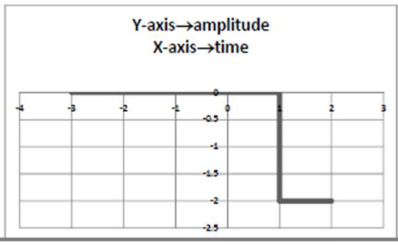
$= 0; t < -1$ $\qquad$ $t = -1$




**03.** $-2u\,(t-1)$

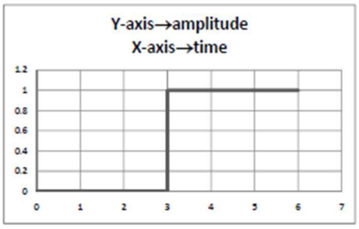
$\therefore\ -2u(t-1) = -2; t \geq 1$ $\qquad$ $here, t-1 = 0$

$= 0;\ t < 1$ $\qquad$ $t = 1$

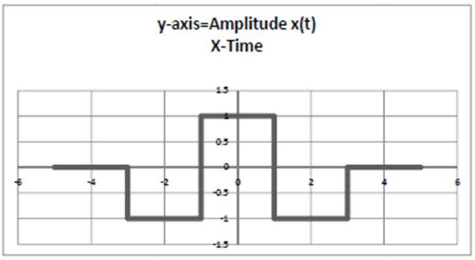


**04.** $u(t-3) =>$ $\qquad\qquad$ $here, t-3=0$

$t=3$




$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$






**Example 91**: Find $L[u(t-a)] = \frac{e^{-as}}{s}$

**Proof: We have**

$L(f(t)) = \int_0^{\infty} f(t)e^{-st}dt$

Here, $f(t) = u(t-a)$

$L(f(t)) = \int_0^{\infty} f(t)e^{-st}dt$

$L(u(t-a)) = \int_0^{\infty} u(t-a)e^{-st}dt$

$L(u(t-a)) = \int_0^{a} 0.e^{-st}dt + \int_a^{\infty} 1.e^{-st}dt$

[The unit function $u(t-a)$ is defined as

$u(t) = 1 \qquad t \geq a$

$\quad = 0 \qquad t < a$

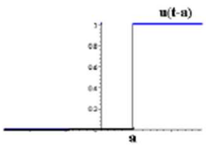



Figure 111

$L(u(t-a)) = \int_0^{a} 0.e^{-st}dt + \int_a^{\infty} 1.e^{-st}dt$

$= 0 + \int_a^{\infty} 1.e^{-st}dt$

$= \int_a^{\infty} e^{-st}dt$

$= \left[\frac{e^{-st}}{-s}\right]_a^{\infty}$

$= -\frac{1}{s}\left[e^{-s\times\infty} - e^{-s\times a}\right]$

$= -\frac{1}{s}\left[e^{-\infty} - e^{-as}\right]$

$$= -\frac{1}{s}\left[\frac{1}{e^{\infty}} - e^{-as}\right]$$

$$= -\frac{1}{s}\left[\frac{1}{\infty} - e^{-as}\right]$$

$$= -\frac{1}{s}\left[0 - e^{-as}\right]$$

$$= +\frac{1}{s}\left[e^{-as}\right]$$

$$= \frac{e^{-as}}{s}$$

$\therefore L(u(t-a)) = \frac{e^{-as}}{s}$ (Proved)

$\therefore L(u(t-2)) = \frac{e^{-2s}}{s}$

**Example 92: Express the following function in terms of unit step functions and find its Laplace transform:**

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

Solution:

**We have**

$L(u(t-a)) = \frac{e^{-as}}{s}$

$\therefore L(u(t-2)) = \frac{e^{-2s}}{s}$ -------------- $(i)$

Given,

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

$$\mathbf{f(t) = \begin{cases} 8+0; & t < 2 \\ 8-2; & t > 2 \end{cases}}$$

$$\mathbf{f(t) = 8 + \begin{cases} 0; & t < 2 \\ -2; & t > 2 \end{cases}}$$

$$\mathbf{f(t) = 8 + (-2)\begin{cases} 0; & t < 2 \\ 1; & t > 2 \end{cases}}$$

$$\mathbf{f(t) = 8 + (-2)\begin{cases} 1; & t > 2 \\ 0; & t < 2 \end{cases}}$$



$\mathbf{f(t) = 8 + (-2)u(t-2)}$
$\mathbf{f(t) = 8 - 2u(t-2)}$
$\mathbf{L\{f(t)\} = L\{8 - 2u(t-2)\}}$
$\mathbf{L\{f(t)\} = L\{8\} - 2Lu(t-2)\}}$
$\mathbf{L\{f(t)\} = 8L\{1\} - 2Lu(t-2)\}}$

$$\mathbf{L\{f(t)\} = 8 \times \frac{1}{s} - 2 \times \frac{e^{-2s}}{s}}$$

$[\because L(1) = \frac{1}{S}$ (*from example* 55)] & $L[u(t-2) = \frac{e^{-2s}}{s}$ (from example 91)]

# Ramp function

**Example 90:**

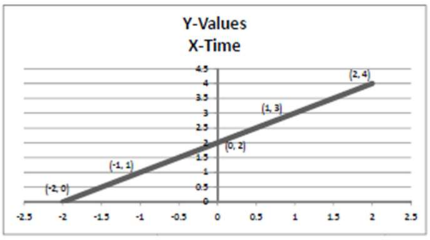
$x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$

Solve:

$r(t+2) = t+2;\ t \geq -2$      *Here,* $t+2=0$

$= 0;\ t < -2$      $\therefore t = -2$

| t | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|
| $r(t+2) = t+2$ | 0 | 1 | 2 | 3 | 4 | 5 |



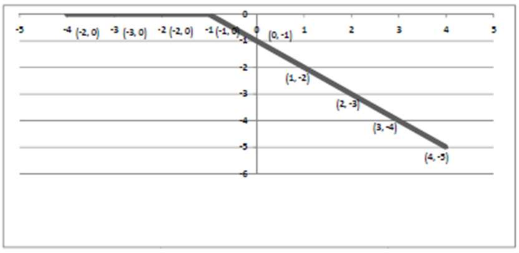

Again,

$-r(t+1) = -(t+1); t \geq -1$      *Here,* $t+1=0$

$= 0;\ t < -1$      $\therefore t = -1$

| $t$ | -1 | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|
| $-r(t+1) = -(t+1)$ | 0 | -1 | -2 | -3 | -4 | -5 |

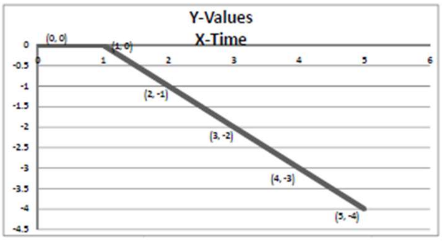





$\therefore -r(t-1) = -(t-1) \quad ; t \geq 1$      *Here,* $t-1=0$

$= 0 \quad ; t < 1$      $\therefore t = 1$

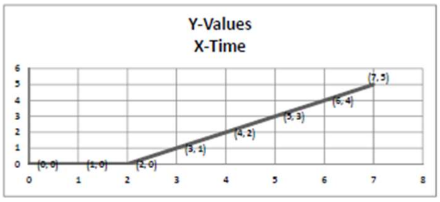
| $t$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $-r(t-1) = -(t-1)$ | 0 | -1 | -2 | -3 | -4 |




$r(t-2) = (t-2) \quad ; t \geq 2$ $\qquad$ $Here, t-2=0$

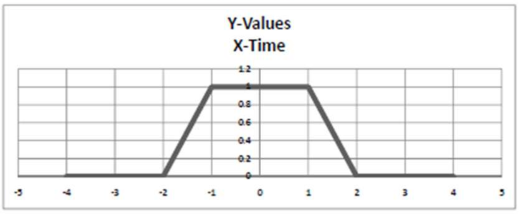
$\quad\;\; = 0 \quad ; t < 2$ $\qquad$ $\therefore t = 2$

| t | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|
| $r(t-2) = t-2$ | 0 | 1 | 2 | 3 | 4 | 5 |






$\therefore x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$




***Problem 31: Impulse Function***

The Impulse function $\delta(t)$ is defined as

$\delta(t) = 1 \qquad t = 0$

$\quad\;\; = 0 \qquad$ Otherwise

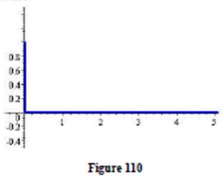
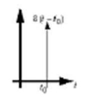


**Figure 110**




Delta (Impulse) Function

## Application to Differential Equations

Q-103: Solve the following Initial Value Problem (IVP) by Laplace Transform: $Y'' + Y = t$, $Y(0) = 1$, $Y'(0) = -2$

Solution

Let $Y = f(t)$

$Y' = f'(t)$

$Y'' = f''(t)$

Given,

$Y'' + Y = t$

That is $\frac{d^2Y}{dt^2} + Y = t$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$Y'' + Y = t$

$L\{Y''\} + L\{Y\} = L\{t\}$

We have,

$L(f''(t)) = s^2 L\{f(t)\} - s\,f(0) - f'(0)$

$L\{Y''\} + L\{Y\} = L\{t\}$

$s^2 L\{Y\} - sf(0) - f'(0) + L\{Y\} = L\{t\}$

$s^2 y - sf(0) - f'(0) + y = L\{t\}$ [let, $L\{Y\} = y$]

$s^2 y - sf(0) - f'(0) + y = \frac{1}{s^2}$ [let, $L\{t\} = \frac{1}{s^2}$]

$s^2 y - s.1 - (-2) + y = \frac{1}{s^2}$ [Given, $Y(0) = f(0) = 1$ $Y'(0) = f'(0) = -2$]

$s^2 y + y - s.1 + 2 = \frac{1}{s^2}$

$s^2 y + y - s.1 + 2 - \frac{1}{s^2} = 0$

$y(s^2+1) - s + 2 - \frac{1}{s^2} = 0$

$y(s^2+1) = s - 2 + \frac{1}{s^2}$

$y = \frac{s-2}{s^2+1} + \frac{1}{s^2(s^2+1)}$

$y = \frac{s}{s^2+1} - \frac{2}{s^2+1} + \frac{1}{s^2} - \frac{1}{(s^2+1)}$

$y = \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$

$\therefore L\{Y\} = y = \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$

$\therefore Y = L^{-1}(y) = L^{-1}[\frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}]$

$\therefore Y = L^{-1}(y) = L^{-1}[\frac{s}{s^2+1}] + L^{-1}[\frac{1}{s^2}] - 3L^{-1}[\frac{1}{s^2+1}]$

$\therefore Y = L^{-1}(y) = \cos t + t - 3\sin t$ Answer

Proof:

$\therefore Y = \cos t + t - 3\sin t$

$\therefore Y' = -\sin t + 1 - 3\cos t$

$\therefore Y'' = -\cos t + 0 + 3\sin t$

$\therefore Y'' + Y = -\cos t + 0 + 3\sin t + \cos t + t - 3\sin t$

$\therefore Y'' + Y = t$

Again,

$\therefore Y = \cos t + t - 3\sin t$

$\therefore Y(0) = \cos 0 + 0 - 3\sin 0$

$\therefore Y(0) = 1$

Again

$\therefore Y' = -\sin t + 1 - 3\cos t$

$\therefore Y'(0) = -\sin 0 + 1 - 3\cos 0$

$\therefore Y'(0) = 0 + 1 - 3.1$

$\therefore Y'(0) = -2$

Q-104: Solve the following Initial Value Problem (IVP) by Laplace Transform $Y'' + 4Y = 12t$

$Y(0) = 0 \qquad Y'(0) = 7$

Solution

Given,

$Y'' + 4Y = 12t$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$Y'' + 4Y = 12t$

$L\{Y''\} + 4L\{Y\} = 12L\{t\}$

We have,

$L(f''(t)) = s^2 L\{f(t)\} - s\,f(0) - f'(0)$

$L\{Y''\} + 4L\{Y\} = 12L\{t\}$

$s^2 L\{Y\} - s f(0) - f'(0) + 4L\{Y\} = 12L\{t\}$

$s^2 y - s f(0) - f'(0) + 4y = 12L\{t\} \qquad [\text{let}, L\{Y\} = y]$

$$s^2y - sf(0) - f'(0) + 4y = 12\frac{1}{s^2} \qquad [\text{let}, L\{t\} = \frac{1}{s^2}]$$

$$s^2y - s.0 - 7 + 4y = 12\frac{1}{s^2} \qquad [\text{Given}, Y(0) = 0 \quad Y'(0) = 7]$$

$$s^2y - 7 + 4y = \frac{12}{s^2}$$

$$s^2y + 4y - 7 - \frac{12}{s^2} = 0$$

$$y(s^2 + 4) - 7 - \frac{12}{s^2} = 0$$

$$y(s^2 + 4) = 7 + \frac{12}{s^2}$$

$$y = \frac{7}{(s^2+4)} + \frac{12}{s^2(s^2+4)}$$

$$y = \frac{7}{(s^2+4)} + 3 \times \frac{4}{s^2(s^2+4)}$$

$$y = \frac{7}{(s^2+4)} + 3 \times [\frac{1}{s^2} - \frac{1}{(s^2+4)}]$$

$$y = \frac{7}{(s^2+4)} + [\frac{3}{s^2} - \frac{3}{(s^2+4)}]$$

$$\therefore L\{Y\} = y = \frac{7}{(s^2+4)} + [\frac{3}{s^2} - \frac{3}{(s^2+4)}]$$

$$\therefore Y = L^{-1}(y) = L^{-1}[\frac{7}{(s^2+4)} + [\frac{3}{s^2} - \frac{3}{(s^2+4)}]]$$

$$\therefore Y = L^{-1}(y) = L^{-1}[\frac{7}{(s^2+4)}] + L^{-1}[\frac{3}{s^2} - \frac{3}{(s^2+4)}]$$

$$\therefore Y = L^{-1}(y) = 7L^{-1}[\frac{1}{(s^2+4)}] + 3L^{-1}[\frac{1}{s^2} - \frac{1}{(s^2+4)}]$$

$$\therefore Y = L^{-1}(y) = 7L^{-1}[\frac{1}{(s^2+4)}] + 3L^{-1}[\frac{1}{s^2}] - 3L^{-1}[\frac{1}{(s^2+4)}]$$

$$\therefore Y = L^{-1}(y) = 4L^{-1}[\frac{1}{(s^2+4)}] + 3L^{-1}[\frac{1}{s^2}]$$

$$\therefore Y = L^{-1}(y) = 4\frac{1}{2}L^{-1}[\frac{2}{(s^2+4)}] + 3L^{-1}[\frac{1}{s^2}]$$

$$\therefore Y = L^{-1}(y) = 4\frac{1}{2}\sin 2t + 3t$$

$$\therefore Y = L^{-1}(y) = 2\sin 2t + 3t$$

Q-105: Solve the following Initial Value Problem (IVP) by Laplace Transform $Y'' - 3Y' + 2Y = 4e^{2t}$

$Y(0) = -3 \qquad Y'(0) = 5$

Solution

$Y = f(t)$

Given,

$Y'' - 3Y' + 2Y = 4e^{2t}$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$Y'' - 3Y' + 2Y = 4e^{2t}$

$L\{Y''\} - 3L\{Y'\} + 2L\{Y\} = 4L\{e^{2t}\}$

We have,

$L(f''(t)) = s^2 L\{f(t)\} - s\,f(0) - f'(0)$

And

$\therefore L(f'(t)) = sL\{f(t)\} - f(0)$

$\therefore L\{Y''\} - 3L\{Y'\} + 2L\{Y\} = 4L\{e^{2t}\}$

$s^2L\{f(t)\} - sf(0) - f'(0) - 3[sL\{f(t)\} - f(0)] + 2y = 4\frac{1}{s-2}$ $\quad$ $[\text{let}, L\{Y\} = y \;\&\; L(e^{at}) = \frac{1}{s-a}]$

$s^2L\{Y\} - sf(0) - f'(0) - 3[sL\{Y\} - f(0)] + 2y = 4\frac{1}{s-2}$ $\quad$ $[Y = f(t)]$

$s^2y - sf(0) - f'(0) - 3[sy - f(0)] + 2y = 4\frac{1}{s-2}$ $\quad$ $[\text{let}, L\{Y\} = y]$

$s^2y - sf(0) - f'(0) - 3sy + 3f(0) + 2y = 4\frac{1}{s-2}$

$s^2y - s\{-3\} - 5 - 3sy + 3(-3) + 2y = 4\frac{1}{s-2}$

$s^2y + 3s - 5 - 3sy - 9 + 2y = 4\frac{1}{s-2}$

$s^2y + 3s - 3sy - 14 + 2y = 4\frac{1}{s-2}$

$s^2y - 3sy + 2y + 3s - 14 = 4\frac{1}{s-2}$

$s^2y - 3sy + 2y = -3s + 14 + 4\frac{1}{s-2}$

$y(s^2 - 3s + 2) = -3s + 14 + 4\frac{1}{s-2}$

$y(s^2 - 2s - s + 2) = -3s + 14 + 4\frac{1}{s-2}$

$y\{s(s-2) - 1(s-2) = -3s + 14 + 4\frac{1}{s-2}$

$$y(s-1)(s-2) = -3s + 14 + 4\frac{1}{s-2}$$

$$y = -3\frac{s}{(s-1)(s-2)} + 14\frac{1}{(s-1)(s-2)} + 4\frac{1}{(s-2)}\frac{1}{(s-1)(s-2)}$$

$$y = \frac{-3s}{(s-1)(s-2)} + \frac{14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2}$$

$$y = \frac{-3s+14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2}$$

$$y = \frac{(-3s+14)(s-2)+4}{(s-1)(s-2)^2}$$

$$y = \frac{-3s^2+6s+14s-28+4}{(s-1)(s-2)^2}$$

$$y = \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2}$$

Applying partial fraction

Let,

$$y = \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-2)^2}$$

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$$y = \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2}$$

$$\therefore L\{Y\} = y = \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2}$$

$$\therefore Y = L^{-1}(y) = -7\frac{1}{2}L^{-1}[\frac{1}{(s-1)}] + 4L^{-1}[\frac{1}{(s-2)}] + 4L^{-1}[\frac{4}{(s-2)^2}]$$

$$\therefore Y = L^{-1}(y) = -7e^{t} + 4e^{2t} + 4te^{2t}$$